Mid-Trade-Amount Normalization — Standalone Factor

Standalone multi-section factor research report

Source: /home/SiYangCao/factor_dev/factor_research0703/factor_dev/research/reports/factors/mid_order_ratio/normalized_v1

Table of Contents

01 — Executive Summary

Frozen five-way decision

Decision: only the fixed-RMB phenomenon remains.

The branch is selected from frozen OOS raw RankIC direction, its two-sided 5% t-stat threshold (|t| >= 1.96 in the frozen direction), and same-direction RankIC in all four PIT pools. Sharpe is shown as a diagnostic and is not used to select the branch. A3 is a secondary P1 diagnostic and does not alter this five-way A1/A2/A0 decision tree.

A0/A1/A2/A3 definitions

Role	Factor ID	Normalization	Frozen bucket
A0	mid_trade_amount_share_abs_4w20w	fixed RMB	(40,000, 200,000] RMB
A1	mid_trade_amount_share_adv20	ADV20 lag-1	(2.0, 20.0] bps of ADV
A2	mid_trade_amount_share_ats20	ATS20 lag-1	(0.50, 2.00] × ATS20
A3	mid_trade_amount_share_rollq	same-day execution quantiles	(Q20, Q80] amount share; P1 diagnostic

All four variants preserve raw-direction RankIC and use frozen effective direction -1 for H-L displays. No window is allowed to re-infer the sign.

CSI1000 full-common-sample metrics

Role	RankIC	ICIR	IC t-stat	H-L Sharpe	H-L MDD	H-L turnover	Coverage
A0	-4.24%	-4.79	-8.91	1.34	-30.12%	1.63	99.99%
A1	2.70%	2.22	4.13	-0.58	-50.16%	1.14	97.73%
A2	-0.58%	-0.54	-1.01	-0.53	-64.38%	1.10	97.73%
A3	-3.60%	-3.42	-6.36	0.71	-40.16%	1.18	100.00%

RankIC and ICIR above are raw-direction statistics. H-L Sharpe, MDD and turnover use the frozen effective direction. Coverage is persisted stock-day factor coverage, not an estimate inserted by this renderer.

The one-way cost label is 7.5 bps, i.e. turnover × 7.5/10,000 × 250 for the display-only implied annual fee. Gross H-L returns remain fee-zero sorting diagnostics.

Four PIT pools

Role	Pool	RankIC	ICIR
A0	ALL	-5.07%	-6.29
A0	CSI300	-2.14%	-2.80
A0	CSI500	-3.04%	-3.28
A0	CSI1000	-4.24%	-4.79
A1	ALL	2.91%	3.11
A1	CSI300	2.14%	1.81
A1	CSI500	2.59%	2.03
A1	CSI1000	2.70%	2.22
A2	ALL	0.14%	0.14
A2	CSI300	-2.96%	-2.42
A2	CSI500	-1.97%	-1.63
A2	CSI1000	-0.58%	-0.54
A3	ALL	-3.57%	-3.98
A3	CSI300	-3.98%	-3.14
A3	CSI500	-3.77%	-3.10

Role	Pool	RankIC	ICIR
A3	CSI1000	-3.60%	-3.42

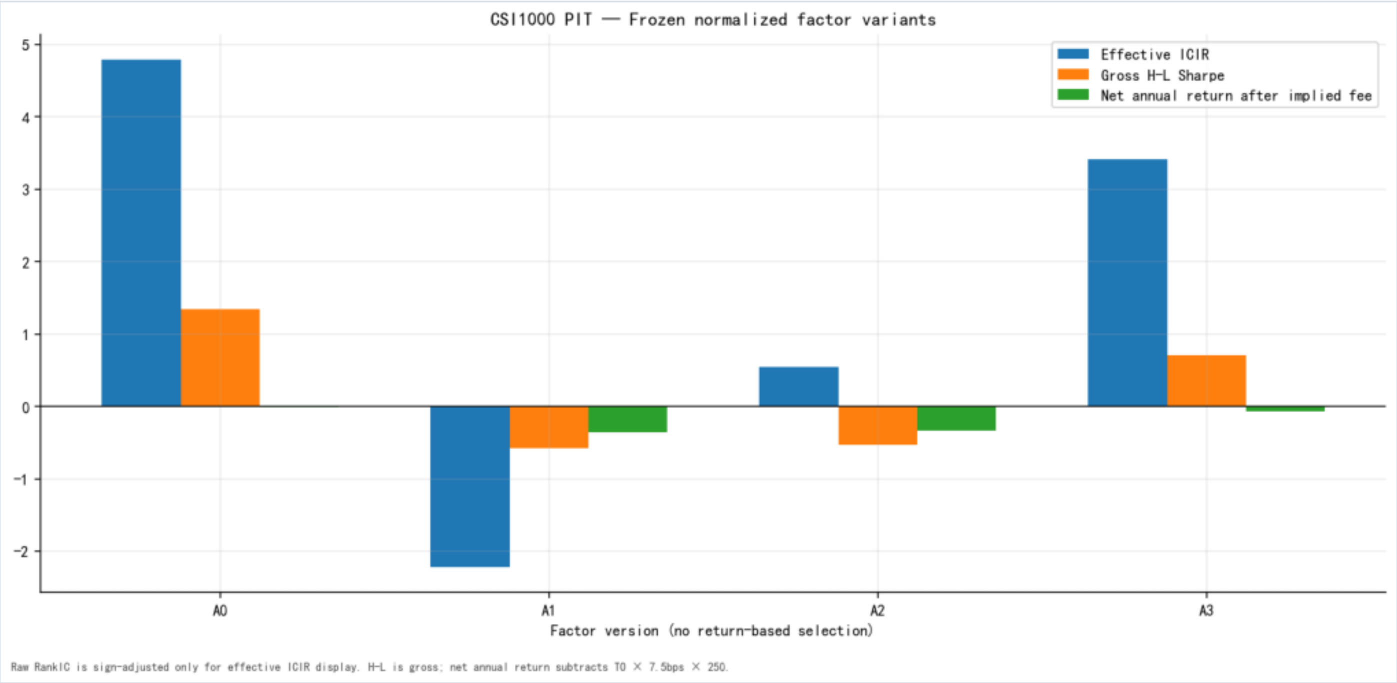
The pools are ALL SSE/SZSE A-shares, CSI300, CSI500 and CSI1000, with point-in-time membership where applicable.

A0 parity gate

- gate: passed;
- Pearson: 1.000000000000;
- Spearman: 1.000000000000;
- maximum absolute error: 1.457e-13;
- mean absolute error: 1.096e-16;
- share within 1e-12: 100.00%.

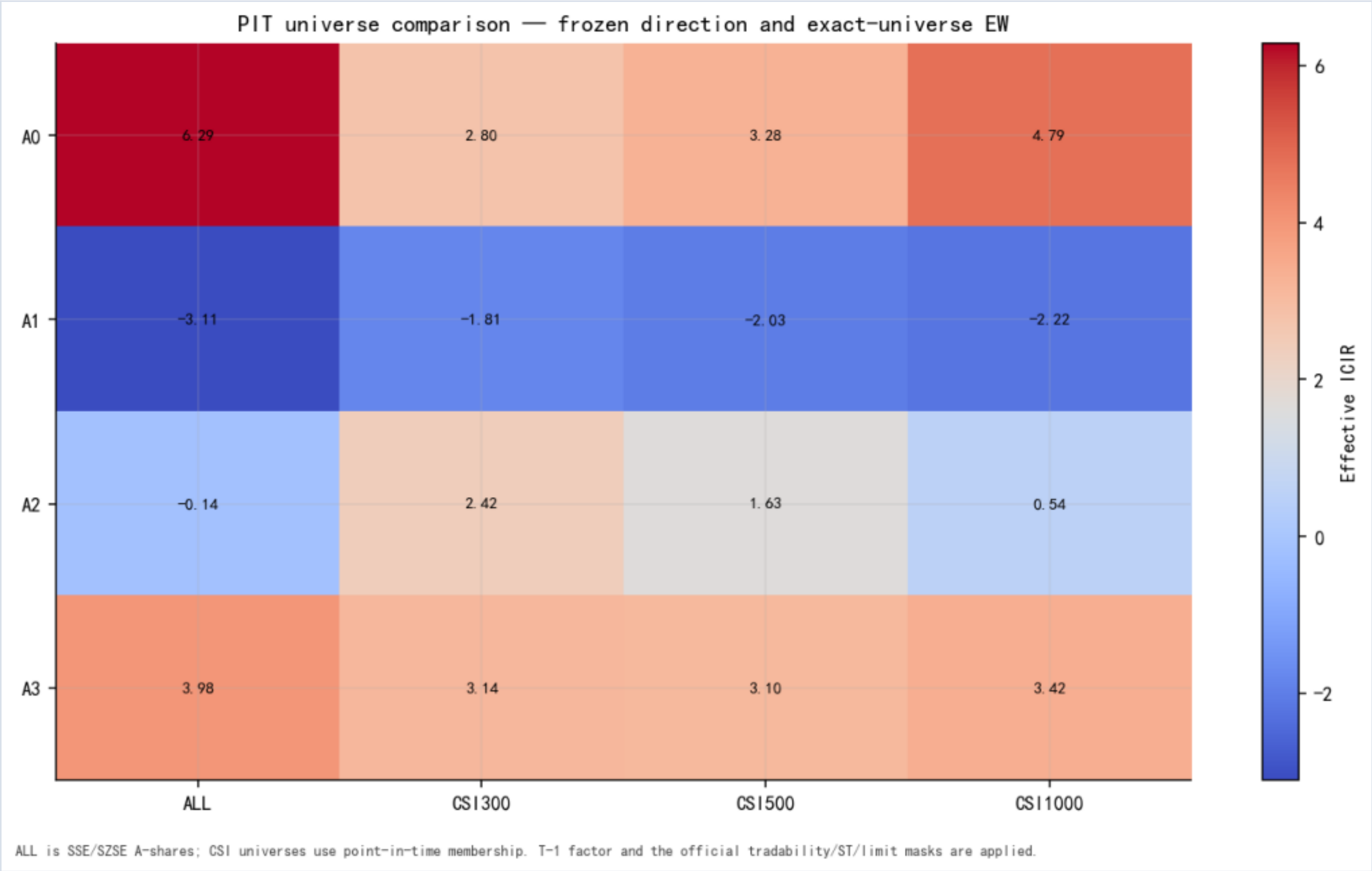
Headline figures

01_factor_variant_summary — Factor variant summary



Factor variant summary — 01_factor_variant_summary

02_universe_variant_summary — PIT universe comparison



PIT universe comparison — 02_universe_variant_summary

Scope boundary

This is standalone single-factor evidence. It does not run factor-library correlation, factor combination, incremental IC, portfolio optimization or alpha stacking.

02 — Problem Definition

What A0 already normalizes

The legacy factor is not completely unnormalized. Its numerator is divided by the stock's same-day total traded amount:

```
A0_{i,t} =
\frac{\sum_k amount_{i,t,k}\,
I(40{,}000 < amount_{i,t,k}\le 200{,}000)}
{\sum_k amount_{i,t,k}}.
```

This removes the mechanical level effect of total daily traded amount from the final ratio.

What A0 does not normalize

The classification boundaries remain fixed at RMB 40,000 and RMB 200,000. They therefore have different economic scale across stocks. A RMB 100,000 execution is 50 bps of ADV for a stock trading RMB 20 million per day, but only 0.5 bp for a stock trading RMB 2 billion per day. A0 classifies both executions identically.

The missing normalization is consequently at the Tick classification boundary, before daily aggregation. It is not the final division by daily turnover.

Why post-factor transforms do not repair it

A daily cross-sectional z-score is a monotone rescaling of already aggregated stock-day values. Industry or market-cap OLS residualization removes linear cross-sectional exposure from the final daily factor. Neither operation can reassign individual executions to a different bucket.

The required order is:

1. normalize every execution amount by a lagged stock-specific scale;
2. classify executions using the normalized scale;
3. aggregate selected execution amounts into a daily amount share;
4. optionally standardize or residualize the daily factor;
5. run standalone predictive validation.

Interpretation boundary

Trade amount is not an investor-identity label. No parent-order reconstruction is performed. The study therefore uses "Tick trade amount," "execution amount," and "trade scale," never "institutional order size" or "retail order size."

03 — Data and Strict Trade Construction

Source records

The source rows are exchange Tick execution records:

- SSE: cmds.SSE_AL_TICK_EXG
- SZSE: cmds.SZSE_AL_TICK_EXG

They are not reconstructed parent orders. Order-number fields on SZSE are used only to identify valid execution records.

Frozen strict definition

For every source row and every trade date:

- session: 09:30:00 <= ExchTime < 15:00:01;
- positive Price and Volume;
- SSE: Type='T';
- SSE amount: ifNull(Amount, Price*Volume);
- SZSE: Type='011' AND BidOrderNo>0 AND AskOrderNo>0;
- SZSE amount: Price*Volume.

Both factor numerators and denominators are sums of these trade amounts, not trade volumes or trade counts.

Server-side construction

Raw Tick rows never cross the network into Python. ClickHouse performs:

1. strict filtering and amount construction;
2. Symbol × TradeDate aggregation;
3. daily amount quantiles and scale primitives;
4. dynamic-threshold sumIf aggregation after joining a small external

stock-day scale table.

Python receives only daily narrow tables. Every cache chunk and combined panel enforces uniqueness on Symbol + TradeDate.

Cache lineage

The normalized cache is separate from the authoritative fixed-bucket cache. Each cache metadata file records:

- SQL text and SHA256;
- strict filters and amount expressions;
- sample dates and symbol scope;
- row count and duplicate-key audit;
- content SHA256;

- ClickHouse client/server versions;
- creation timestamp and chunk lineage.

The existing strict fixed-bucket cache is never overwritten.

04 — Normalized Factor Definitions

A0 — Fixed RMB benchmark

```
mid\_trade\_amount\_share\_abs\_4w20w_{i,t} =
\frac{\sum_k amount_{i,t,k}
I(40{,}000 < amount_{i,t,k}\le 200{,}000)}
{\sum_k amount_{i,t,k}}.
```

mid_order_ratio is retained only as a legacy alias. A0 is the parity benchmark and is not described as cross-stock scale normalized.

A1 — Lagged ADV20 normalization

```
ADV20^{lag1}_{i,t}
= mean(TotalAmount_{i,t-20},\ldots,TotalAmount_{i,t-1}),
```

```
relative\_trade\_size^{ADV}_{i,t,k}
= \frac{amount_{i,t,k}}{ADV20^{lag1}_{i,t}}\times 10{,}000
\quad\text{bps of ADV},
```

```
A1_{i,t} =
\frac{\sum_k amount_{i,t,k}
I(L_{adv}<relative\_trade\_size^{ADV}_{i,t,k}\le H_{adv})}
{\sum_k amount_{i,t,k}}.
```

The main bounds are frozen from the 2023-01-03 to 2023-06-30 trade-size distribution without loading returns. The candidate grid is $L=\{0.5,1,2\}$ bps and $H=\{5,10,20\}$ bps. The deterministic selection matches A0's amount coverage subject to non-sparse market-cap-quintile coverage.

A2 — Lagged typical Tick amount

For each stock-day, ClickHouse first computes the median positive Tick execution amount. The historical scale is:

```
ATS20^{lag1}_{i,t}
= median(DailyMedianTradeAmount_{i,t-20},\ldots,
DailyMedianTradeAmount_{i,t-1}).
```

```
A2_{i,t} =
\frac{\sum_k amount_{i,t,k}
I(0.5 < amount_{i,t,k}/ATS20^{lag1}_{i,t}\le 2.0)}
{\sum_k amount_{i,t,k}}.
```

The main 0.5–2.0 interval is frozen ex ante. The $L=\{0.25,0.5,0.75\}$, $H=\{1.5,2.0,3.0\}$ grid is a stability diagnostic, not a parameter-selection exercise.

A3 — Current-day relative quantiles

```
A3_{i,t} =
\frac{\sum_k amount_{i,t,k}
I(Q20_{i,t}<amount_{i,t,k}\le Q80_{i,t})}
```

```
{\sum_k amount_{i,t,k}}.
```

A3 is known only after the close and is shifted one trading day before evaluation. It describes the middle of the current day's execution-amount distribution, not deviation from a historical norm. Because count coverage would be partly mechanical, the factor remains an amount share.

No-look-ahead and missing history

ADV20 and ATS20 use the previous 20 market trading dates after reindexing to the complete trading calendar. A missing source day makes the scale and factor NaN; older observations are not pulled forward to fill the window. Every valid scale row stores a maximum source date no later than t-1.

All variants use raw-direction RankIC. Effective-direction portfolio diagnostics use the ex-ante frozen direction -1; no evaluation window may re-infer the sign.

05 — Standalone Validation

CSI1000 standalone and decile diagnostics

Role	RankIC	ICIR	t-stat	H-L ann. return	Sharpe	MDD	Turnover	Decile mono.	Index-excess H-L	Decile turnover	Implied fee	Coverage
A0	-4.24%	-4.79	-8.91	30.20%	1.34	-30.12%	1.63	0.830	30.20%	1.63	30.59%	99.99%
A1	2.70%	2.22	4.13	-14.52%	-0.58	-50.16%	1.14	-0.527	-14.52%	1.14	21.42%	97.73%
A2	-0.58%	-0.54	-1.01	-12.05%	-0.53	-64.38%	1.10	-0.309	-12.05%	1.10	20.69%	97.73%
A3	-3.60%	-3.42	-6.36	15.88%	0.71	-40.16%	1.18	0.952	15.88%	1.18	22.10%	100.00%

Implied fee is the persisted display-only annual deduction under a one-way 7.5 bps label. It is not a claim that the gross H-L sort is a deployable strategy. RankIC, ICIR, Sharpe, MDD, turnover and coverage are reported together so that a high Sharpe cannot hide weak information coefficient evidence or sparse coverage.

Four-pool validation

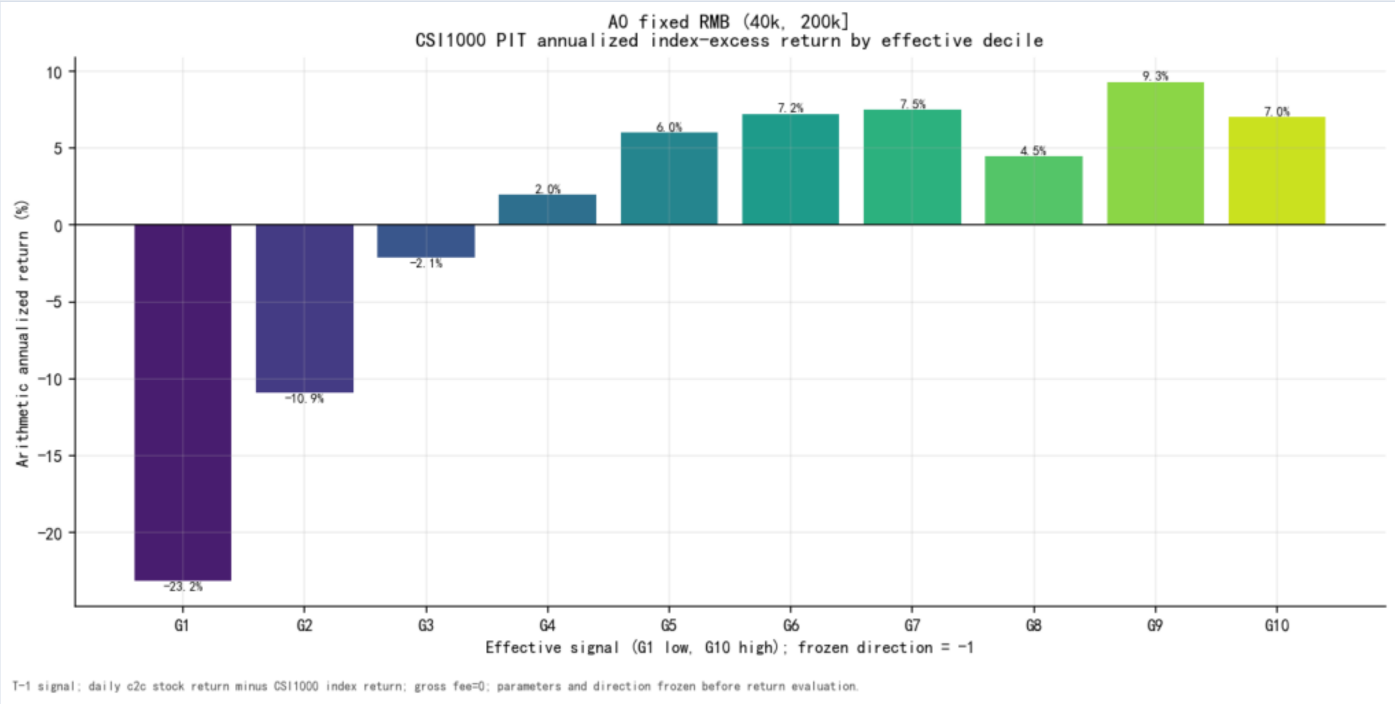
Role	Pool	RankIC	t-stat	ICIR	Sharpe	MDD	Turnover
A0	ALL	-5.07%	-11.70	-6.29	2.20	-25.35%	1.69
A0	CSI300	-2.14%	-5.20	-2.80	0.55	-32.30%	1.58
A0	CSI500	-3.04%	-6.10	-3.28	0.48	-37.35%	1.65
A0	CSI1000	-4.24%	-8.91	-4.79	1.34	-30.12%	1.63
A1	ALL	2.91%	5.79	3.11	-1.38	-63.67%	1.09
A1	CSI300	2.14%	3.37	1.81	0.14	-37.25%	1.13
A1	CSI500	2.59%	3.77	2.03	-0.18	-41.75%	1.15
A1	CSI1000	2.70%	4.13	2.22	-0.58	-50.16%	1.14
A2	ALL	0.14%	0.27	0.14	-0.85	-62.85%	1.16
A2	CSI300	-2.96%	-4.50	-2.42	0.39	-46.47%	0.94

Role	Pool	RankIC	t-stat	ICIR	Sharpe	MDD	Turnover
A2	CSI500	-1.97%	-3.04	-1.63	-0.06	-57.45%	0.99
A2	CSI1000	-0.58%	-1.01	-0.54	-0.53	-64.38%	1.10
A3	ALL	-3.57%	-7.40	-3.98	1.34	-26.12%	1.26
A3	CSI300	-3.98%	-5.83	-3.14	0.51	-34.55%	1.03
A3	CSI500	-3.77%	-5.77	-3.10	0.32	-44.89%	1.04
A3	CSI1000	-3.60%	-6.36	-3.42	0.71	-40.16%	1.18

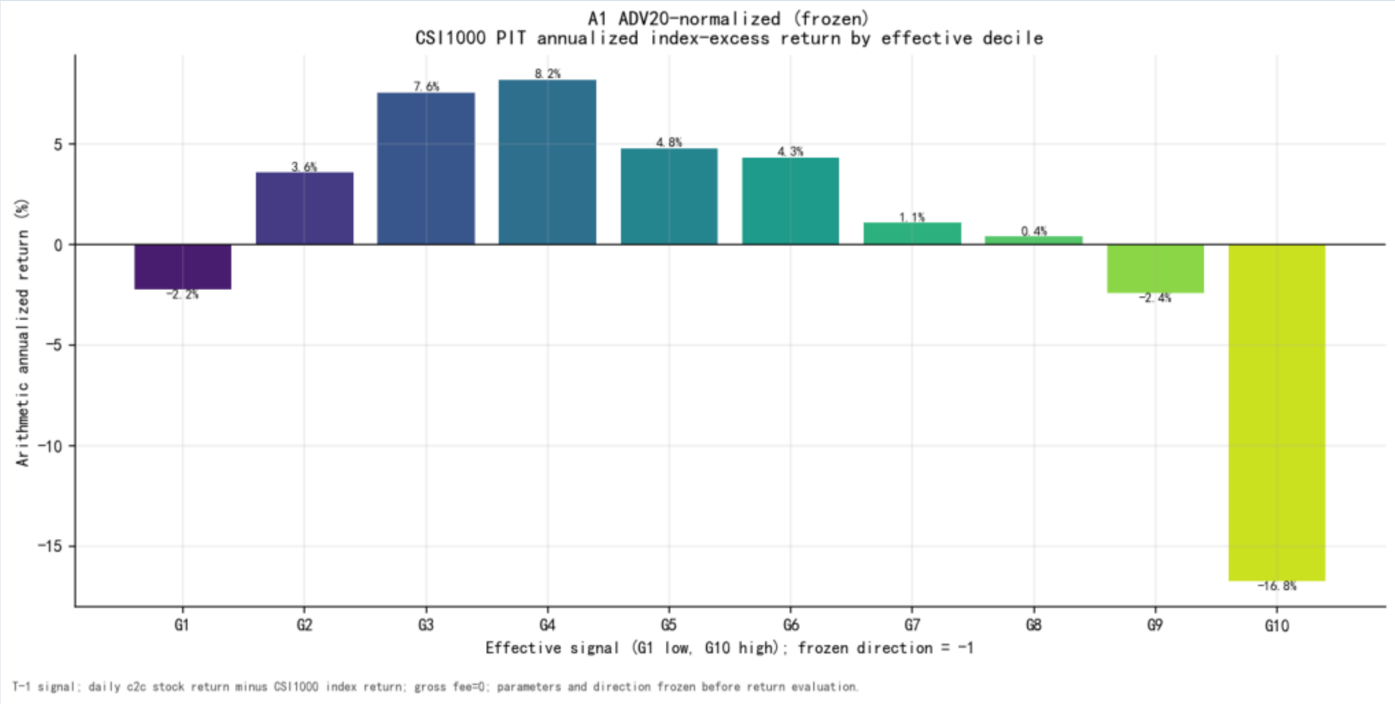
The four-pool table uses exact valid-universe equal weighting. Cross-pool direction is a decision input; the maximum Sharpe across pools is not.

Per-variant decile figures

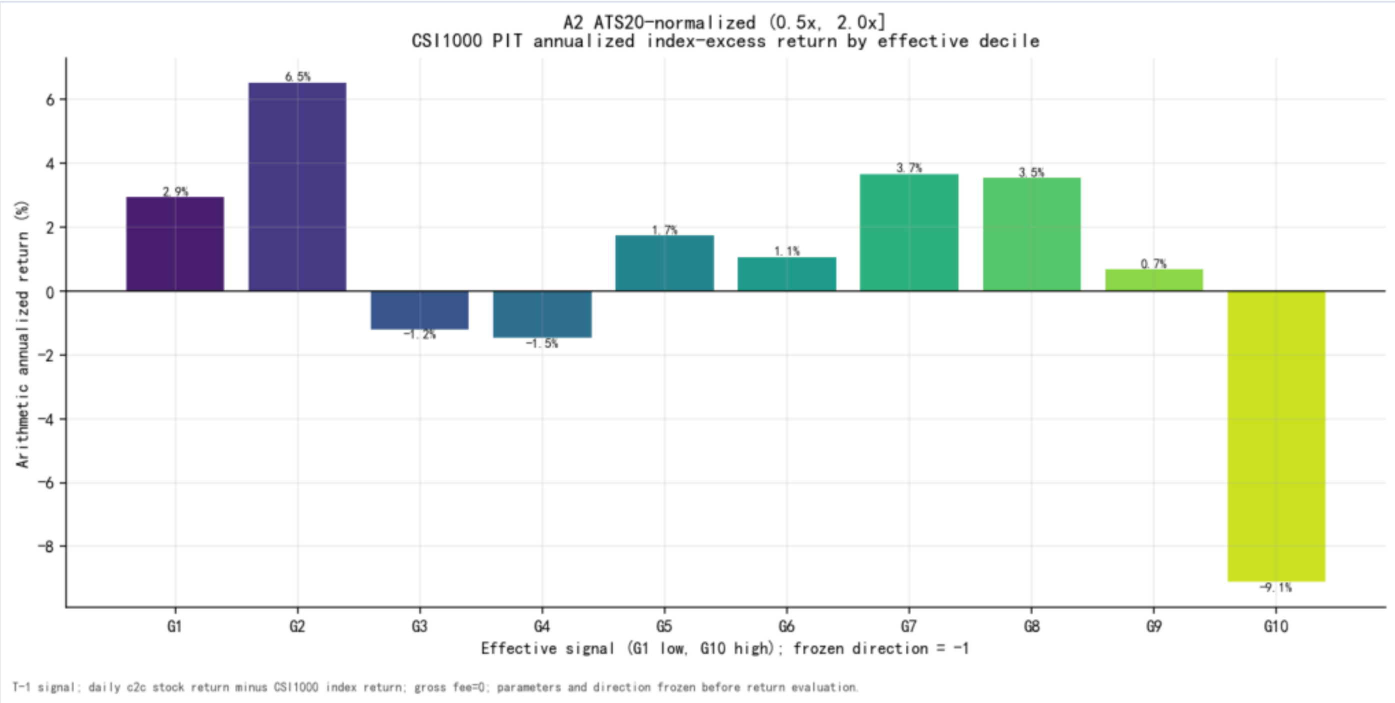
03_decile_annualized — Per-variant decile annualized return



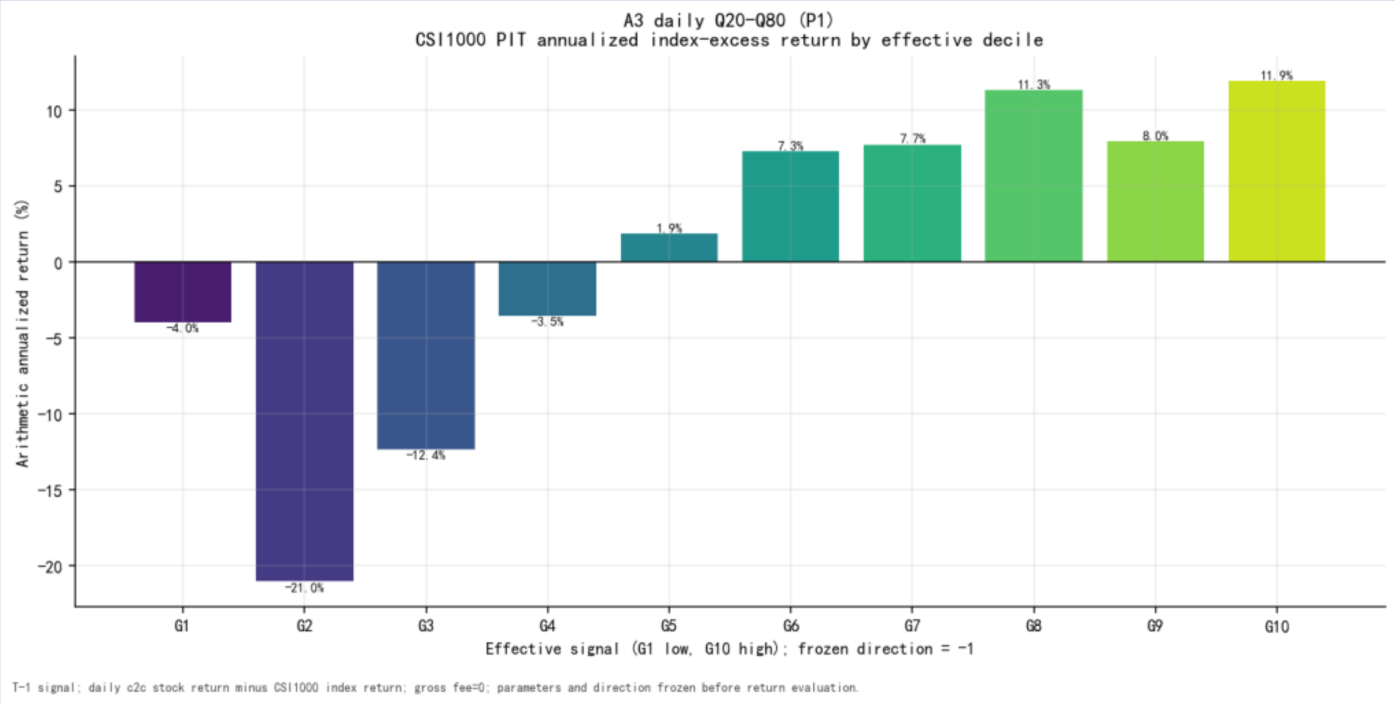
Per-variant decile annualized return — 03_decile_annualized__mid_trade_amount_share_abs_4w20w



Per-variant decile annualized return — 03_decile_annualized__mid_trade_amount_share_adv20

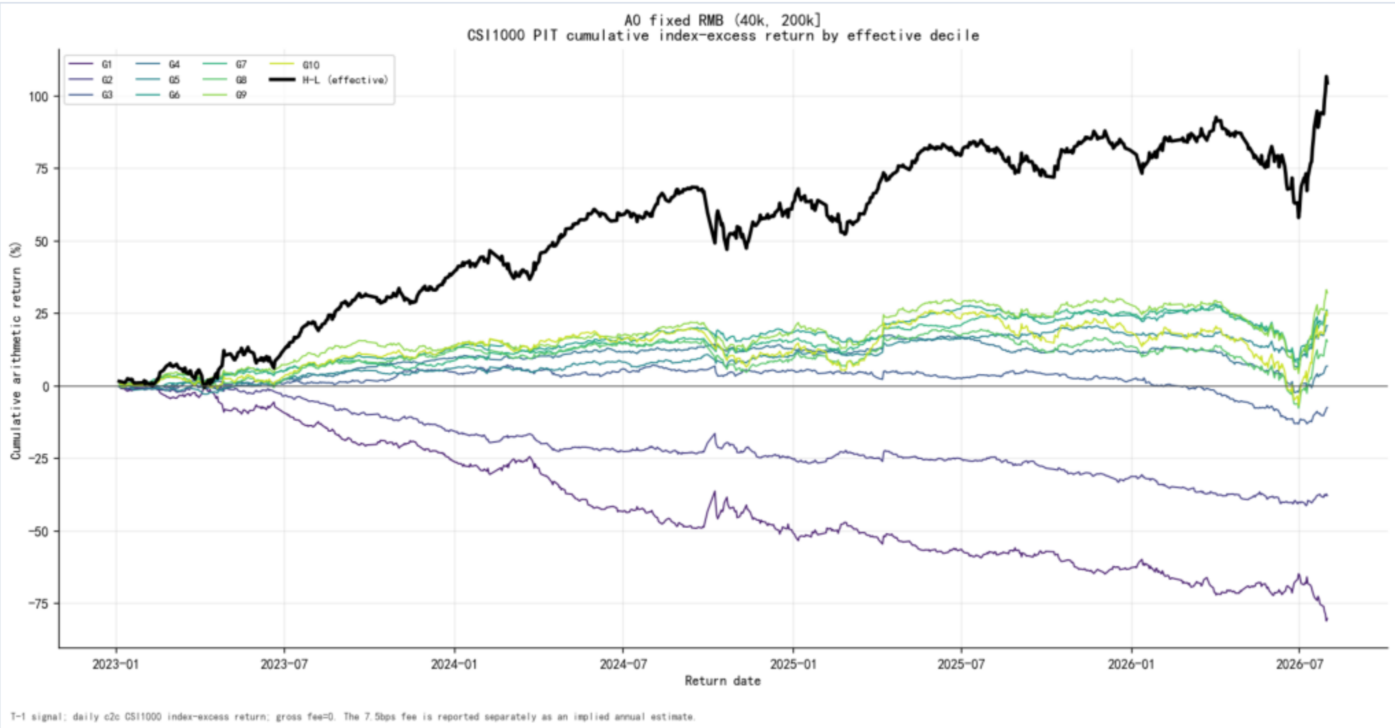


Per-variant decile annualized return — 03_decile_annualized__mid_trade_amount_share_ats20

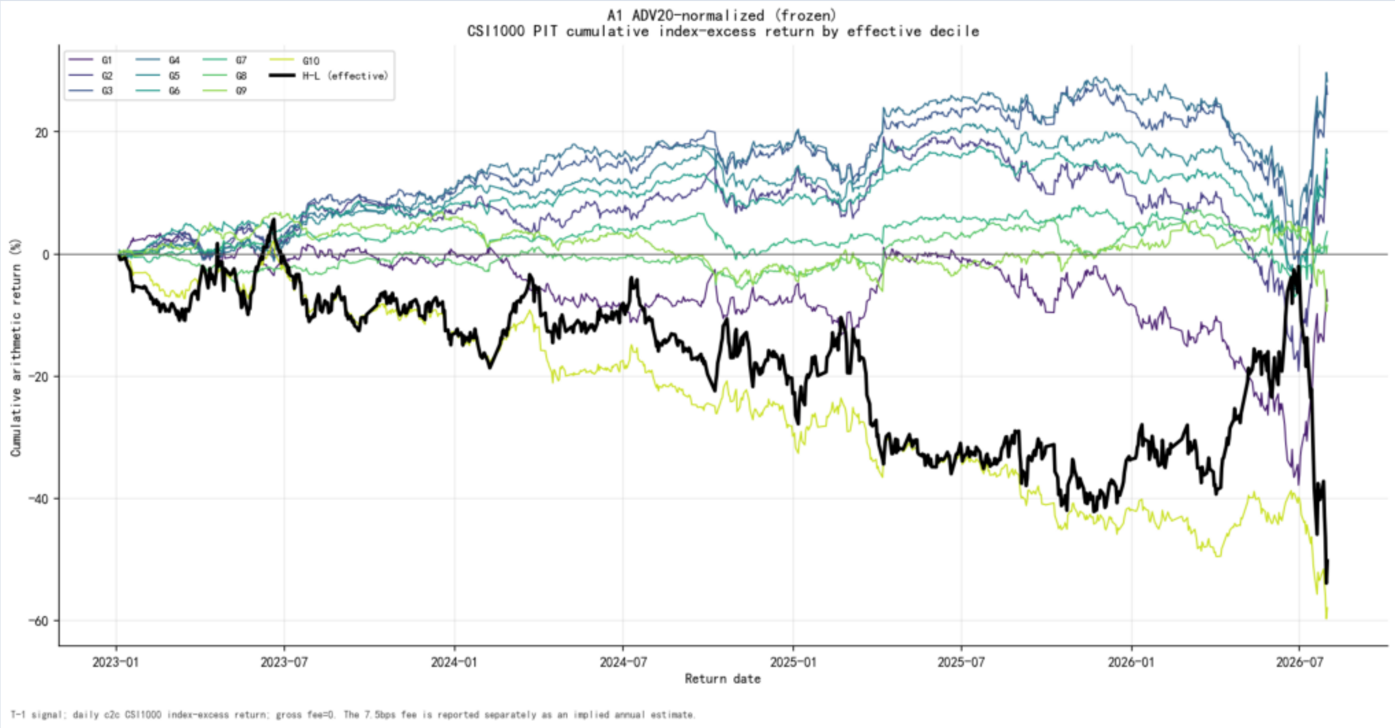


Per-variant decile annualized return — 03_decile_annualized__mid_trade_amount_share_rollq

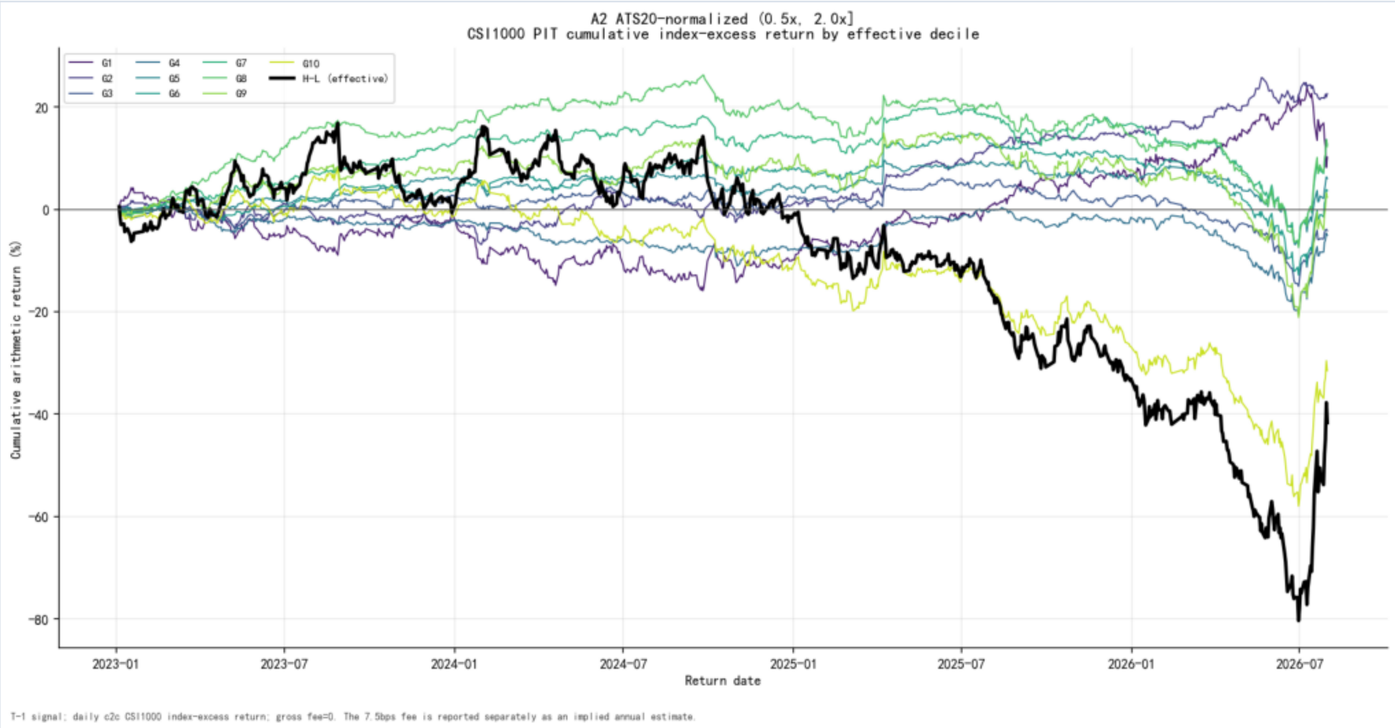
04_decile_cumulative — Per-variant decile cumulative return



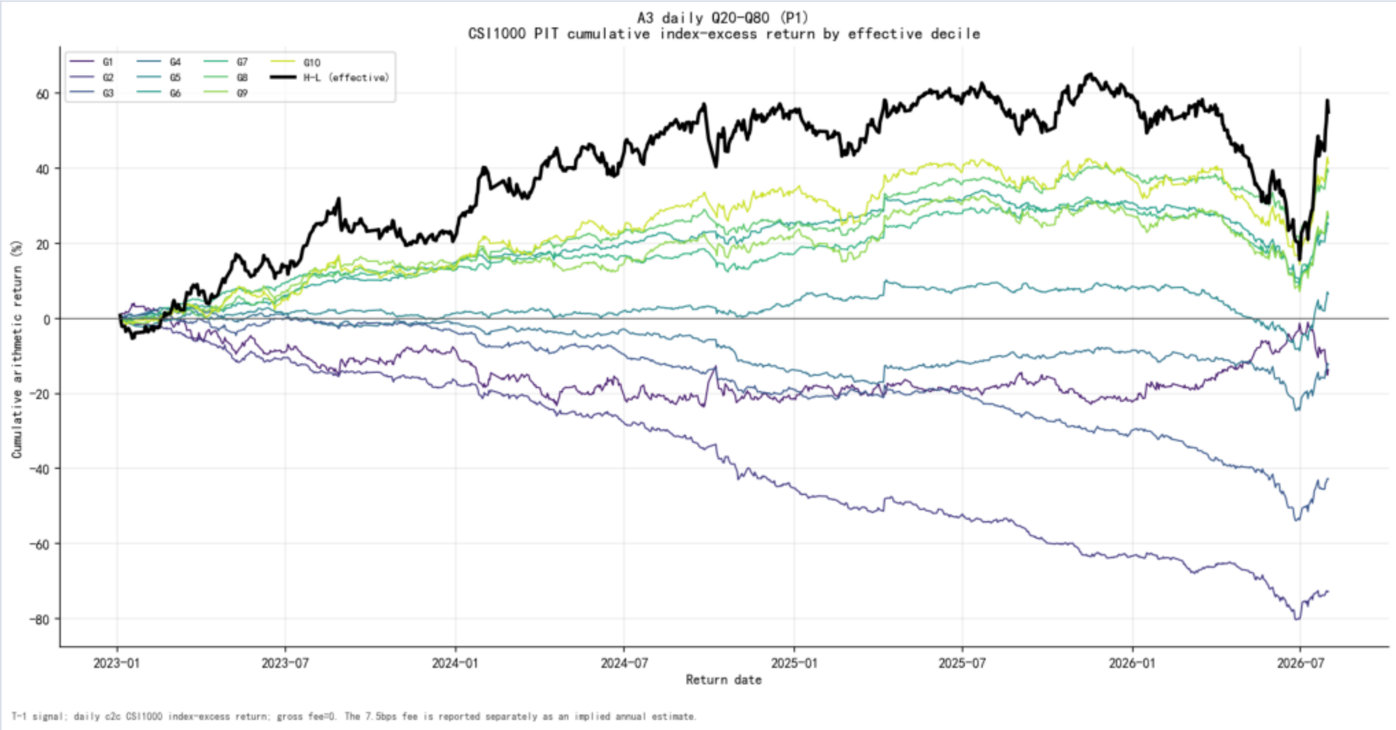
Per-variant decile cumulative return — 04_decile_cumulative__mid_trade_amount_share_abs_4w20w



Per-variant decile cumulative return — 04_decile_cumulative__mid_trade_amount_share_adv20



Per-variant decile cumulative return — 04_decile_cumulative__mid_trade_amount_share_ats20



Per-variant decile cumulative return — 04_decile_cumulative__mid_trade_amount_share_rollq

06 — Normalization Diagnostics

Frozen A1 distribution calibration

The A1 bounds were selected from the 2023-H1 execution-size distribution, without return data. Values below are persisted bps-of-ADV20 observations.

Quantiles across four pools

Pool	Quantile	bps of ADV
ALL	0.10	0.0277
ALL	0.25	0.0666
ALL	0.50	0.1822
ALL	0.75	0.5272
ALL	0.90	1.4415
ALL	0.95	2.6870
ALL	0.99	8.8401
CSI1000	0.10	0.0319
CSI1000	0.25	0.0715
CSI1000	0.50	0.1809
CSI1000	0.75	0.4931
CSI1000	0.90	1.2817
CSI1000	0.95	2.3138
CSI1000	0.99	7.0045
CSI300	0.10	0.0129

Pool	Quantile	bps of ADV
CSI300	0.25	0.0315
CSI300	0.50	0.0813
CSI300	0.75	0.2144
CSI300	0.90	0.5245
CSI300	0.95	0.9101
CSI300	0.99	2.7027
CSI500	0.10	0.0239
CSI500	0.25	0.0519
CSI500	0.50	0.1330
CSI500	0.75	0.3562
CSI500	0.90	0.9028
CSI500	0.95	1.6117
CSI500	0.99	4.9388

Median by CSI1000 market-cap quintile

Cap quintile	Median bps of ADV
Q1	0.2488
Q2	0.2117
Q3	0.1995
Q4	0.1780
Q5	0.1325

ADV candidate coverage grid

L bps	H bps	Frozen	Coverage	A0 coverage	\	diff\		Mean cap-Q \	diff\		Min cap-Q	Max cap-Q
2.0	20.0	yes	31.03%	31.92%	0.89%	6.32%	24.39%	41.53%				
1.0	5.0	no	32.97%	31.92%	1.05%	5.10%	29.05%	37.19%				
2.0	10.0	no	25.85%	31.92%	6.07%	5.10%	20.82%	33.46%				
1.0	10.0	no	41.61%	31.92%	9.69%	12.72%	35.45%	49.63%				
2.0	5.0	no	17.21%	31.92%	14.71%	12.90%	14.42%	21.02%				
1.0	20.0	no	46.79%	31.92%	14.87%	18.58%	39.02%	57.70%				
0.5	5.0	no	48.41%	31.92%	16.49%	18.35%	44.85%	51.19%				
0.5	10.0	no	57.06%	31.92%	25.14%	27.91%	51.25%	63.41%				
0.5	20.0	no	62.24%	31.92%	30.32%	33.77%	54.82%	71.48%				

The frozen row is selected by distribution coverage relative to A0, subject to the persisted 10%–80% cap-quintile gate. Return, RankIC and Sharpe are not inputs to this calibration.

Frozen A1 coverage by market-cap bucket

Bucket type	Bucket	A1 coverage	A0 coverage
market_cap	1	41.53%	29.05%
market_cap	2	36.18%	30.27%
market_cap	3	34.53%	30.94%
market_cap	4	31.75%	31.65%
market_cap	5	24.39%	33.88%

Missing-scale and factor coverage

Role	Required scale	Expected stock-days	Factor stock-days	Coverage	Missing scale	Coverage given scale
A0	none	4,413,448	4,413,099	99.99%	0.00%	99.99%
A1	adv20_lag1	4,413,448	4,313,073	97.73%	2.27%	100.00%
A2	ats20_lag1	4,413,448	4,313,073	97.73%	2.27%	100.00%
A3	none	4,413,448	4,413,448	100.00%	0.00%	100.00%

ADV20 and ATS20 require exactly 20 lagged market trading dates. Missing history remains missing; the renderer does not fill or extrapolate it.

Return-side parameter stability

Role	Factor ID	Bounds	Unit	Selected	RankIC	ICIR	t-stat
A0	mid_trade_amount_share_abs_4w20w	(40000.00, 200000.00]	RMB_per_trade	yes	-4.24%	-4.79	-8.91
A1	a1_adv20_l0p5_h5_bps	(0.50, 5.00]	bps_of_ADV20_lag1	no	-0.52%	-0.56	-1.04
A1	a1_adv20_l0p5_h10_bps	(0.50, 10.00]	bps_of_ADV20_lag1	no	0.56%	0.54	1.01
A1	a1_adv20_l0p5_h20_bps	(0.50, 20.00]	bps_of_ADV20_lag1	no	1.36%	1.22	2.26
A1	a1_adv20_l1_h5_bps	(1.00, 5.00]	bps_of_ADV20_lag1	no	0.40%	0.39	0.72
A1	a1_adv20_l1_h10_bps	(1.00, 10.00]	bps_of_ADV20_lag1	no	1.45%	1.28	2.37
A1	a1_adv20_l1_h20_bps	(1.00, 20.00]	bps_of_ADV20_lag1	no	2.06%	1.74	3.23
A1	a1_adv20_l2_h5_bps	(2.00, 5.00]	bps_of_ADV20_lag1	no	1.44%	1.30	2.42
A1	a1_adv20_l2_h10_bps	(2.00, 10.00]	bps_of_ADV20_lag1	no	2.30%	1.94	3.60
A1	a1_adv20_l2_h20_bps	(2.00, 20.00]	bps_of_ADV20_lag1	no	2.70%	2.22	4.13

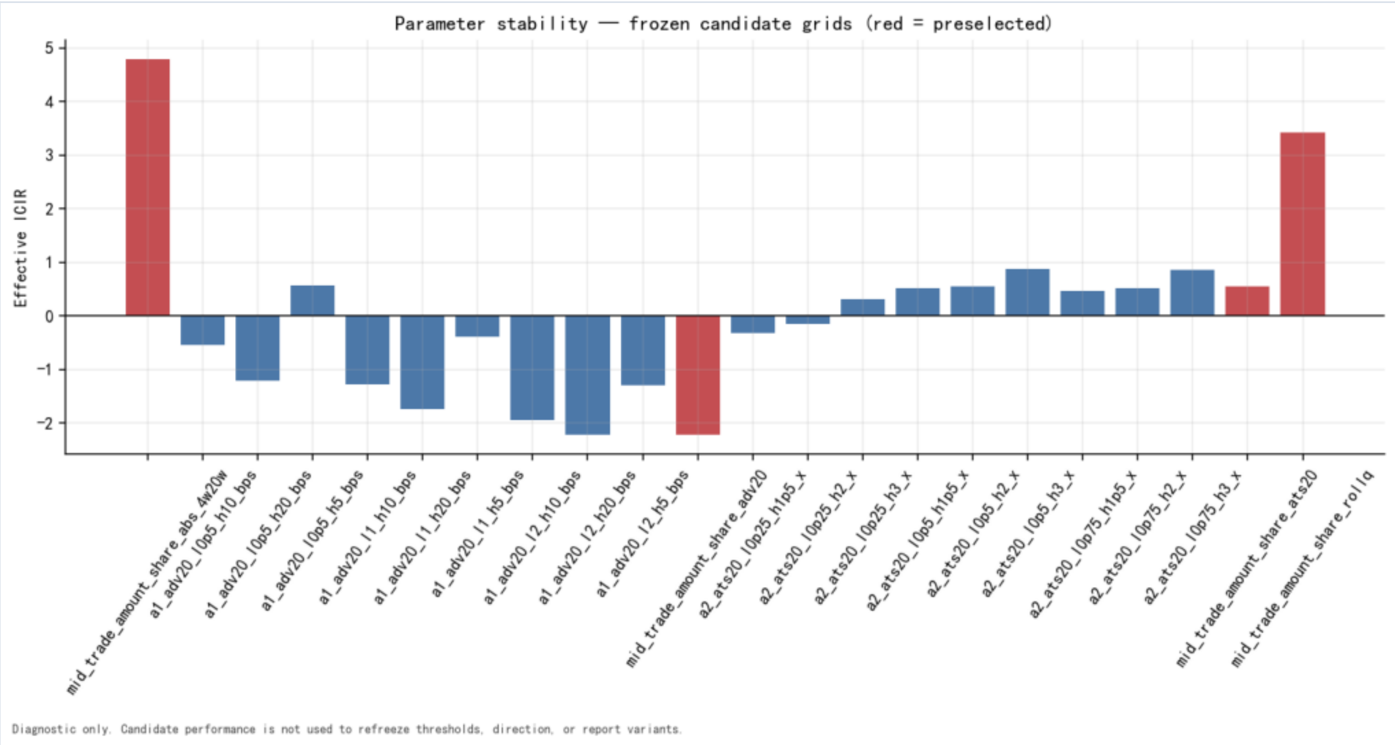
Role	Factor ID	Bounds	Unit	Selected	RankIC	ICIR	t-stat
A1	mid_trade_amount_share_adv20	(2.00, 20.00]	bps_of_ADV20_lag1	yes	2.70%	2.22	4.13
A2	a2_ats20_l0p25_h1p5_x	(0.25, 1.50]	multiple_of_ATS20_lag1	no	0.34%	0.33	0.61
A2	a2_ats20_l0p25_h2_x	(0.25, 2.00]	multiple_of_ATS20_lag1	no	0.15%	0.14	0.27
A2	a2_ats20_l0p25_h3_x	(0.25, 3.00]	multiple_of_ATS20_lag1	no	−0.32%	−0.31	−0.57
A2	a2_ats20_l0p5_h1p5_x	(0.50, 1.50]	multiple_of_ATS20_lag1	no	−0.55%	−0.51	−0.95
A2	a2_ats20_l0p5_h2_x	(0.50, 2.00]	multiple_of_ATS20_lag1	no	−0.58%	−0.54	−1.01
A2	mid_trade_amount_share_ats20	(0.50, 2.00]	multiple_of_ATS20_lag1	yes	−0.58%	−0.54	−1.01
A2	a2_ats20_l0p5_h3_x	(0.50, 3.00]	multiple_of_ATS20_lag1	no	−0.93%	−0.87	−1.62
A2	a2_ats20_l0p75_h1p5_x	(0.75, 1.50]	multiple_of_ATS20_lag1	no	−0.51%	−0.47	−0.88
A2	a2_ats20_l0p75_h2_x	(0.75, 2.00]	multiple_of_ATS20_lag1	no	−0.56%	−0.52	−0.96

Role	Factor ID	Bounds	Unit	Selected	RankIC	ICIR	t-stat
A2	a2_ats20_l0p75_h3_x	(0.75, 3.00]	multiple_of_ATS20_lag1	no	−0.93%	−0.86	−1.59
A3	mid_trade_amount_share_rollq	(0.20, 0.80]	same_day_trade_amount_quantile	yes	−3.60%	−3.42	−6.36

This table is diagnostic after freezing. The selected flags and all candidate metrics are read from parameter_stability.csv; no candidate is selected by the best Sharpe or ICIR.

Parameter-stability figure

07_parameter_stability — Frozen-grid parameter stability



Frozen-grid parameter stability — 07_parameter_stability

07 — Exposure Diagnostics

CSI1000 market-cap quintiles

Role	Cap quintile	RankIC	ICIR	Coverage	Avg names	Days
A0	Q1	-3.91%	-4.83	100.00%	196.8	865
A0	Q2	-3.98%	-4.67	100.00%	197.2	865
A0	Q3	-3.97%	-4.48	100.00%	197.2	865
A0	Q4	-3.99%	-4.14	100.00%	197.2	865
A0	Q5	-3.82%	-3.34	100.00%	197.6	865
A1	Q1	0.90%	0.83	100.00%	194.0	865
A1	Q2	1.67%	1.42	100.00%	194.5	865
A1	Q3	2.12%	1.71	100.00%	194.4	865
A1	Q4	2.52%	1.98	100.00%	194.5	865
A1	Q5	2.76%	2.19	100.00%	194.8	865
A2	Q1	-0.00%	-0.00	100.00%	194.0	865
A2	Q2	0.00%	0.00	100.00%	194.5	865
A2	Q3	-0.23%	-0.21	100.00%	194.4	865
A2	Q4	-0.39%	-0.38	100.00%	194.5	865
A2	Q5	0.05%	0.04	100.00%	194.8	865

Role	Cap quintile	RankIC	ICIR	Coverage	Avg names	Days
A3	Q1	-3.31%	-2.88	100.00%	196.8	865
A3	Q2	-3.67%	-3.21	100.00%	197.2	865
A3	Q3	-3.42%	-2.96	100.00%	197.2	865
A3	Q4	-3.21%	-3.13	100.00%	197.2	865
A3	Q5	-2.32%	-2.19	100.00%	197.6	865

CSI1000 lagged-ADV quintiles

Role	ADV quintile	RankIC	ICIR	Coverage	Avg names	Days
A0	Q1	-1.96%	-2.88	98.61%	194.0	865
A0	Q2	-1.57%	-2.31	98.61%	194.5	865
A0	Q3	-2.01%	-2.82	98.61%	194.4	865
A0	Q4	-2.78%	-3.66	98.61%	194.5	865
A0	Q5	-4.19%	-5.04	98.61%	194.8	865
A1	Q1	-2.15%	-3.10	100.00%	194.0	865
A1	Q2	-1.92%	-2.78	100.00%	194.5	865
A1	Q3	-1.64%	-2.26	100.00%	194.4	865
A1	Q4	-1.50%	-2.06	100.00%	194.5	865
A1	Q5	1.27%	1.53	100.00%	194.8	865
A2	Q1	-0.20%	-0.16	100.00%	194.0	865
A2	Q2	-1.45%	-1.10	100.00%	194.5	865
A2	Q3	-0.67%	-0.58	100.00%	194.4	865
A2	Q4	0.15%	0.15	100.00%	194.5	865
A2	Q5	1.48%	1.52	100.00%	194.8	865

Role	ADV quintile	RankIC	ICIR	Coverage	Avg names	Days
A3	Q1	-2.49%	-1.97	98.61%	194.0	865
A3	Q2	-3.66%	-2.94	98.61%	194.5	865
A3	Q3	-3.27%	-2.85	98.61%	194.4	865
A3	Q4	-2.95%	-2.90	98.61%	194.5	865
A3	Q5	-3.25%	-3.61	98.61%	194.8	865

Both characteristic sorts are daily cross-sectional quintiles. The tables show actual per-stratum coverage and name counts; a missing stratum is a hard input failure rather than an omitted row.

OLS diagnostics: raw / industry / cap / joint

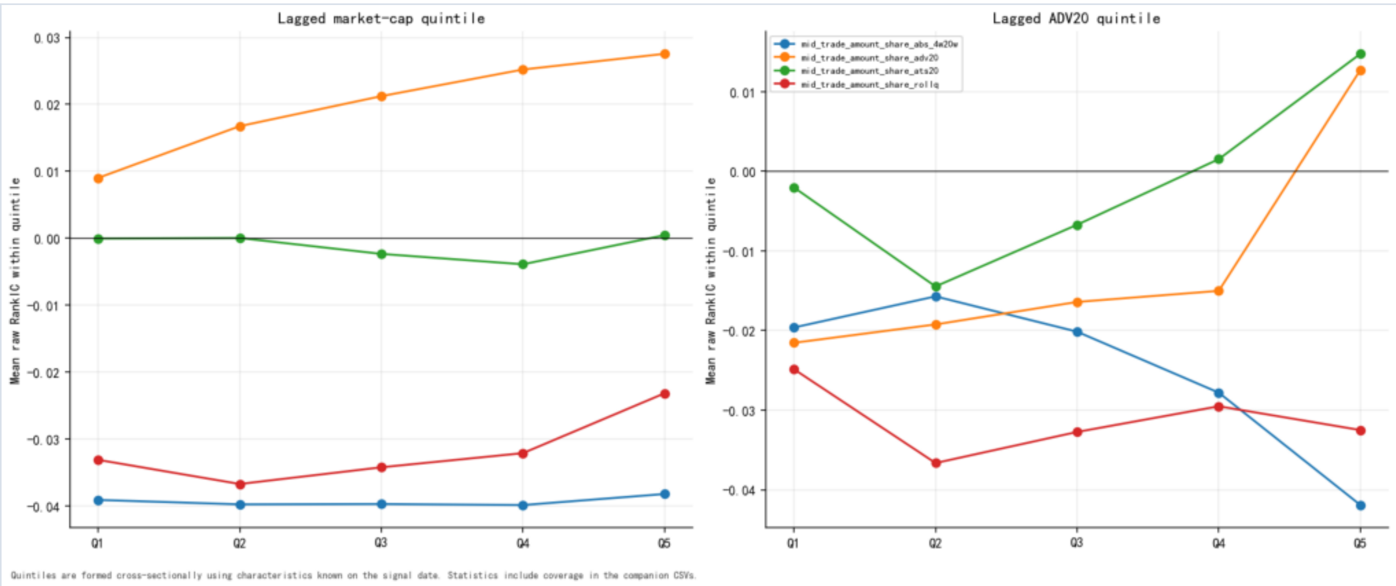
Role	OLS method	RankIC	ICIR	t-stat	\	RankIC\	retained
A0	raw	-4.24%	-4.79	-8.91	100.00%		
A0	industry	-4.00%	-7.82	-14.54	94.21%		
A0	cap	-3.67%	-4.62	-8.60	86.48%		
A0	joint	-3.56%	-7.25	-13.48	84.04%		
A1	raw	2.70%	2.22	4.13	100.00%		
A1	industry	2.33%	3.13	5.81	86.00%		
A1	cap	2.05%	1.84	3.43	75.85%		
A1	joint	1.76%	2.60	4.84	65.08%		
A2	raw	-0.58%	-0.54	-1.01	100.00%		
A2	industry	-0.17%	-0.28	-0.52	28.75%		

Role	OLS method	RankIC	ICIR	t-stat	\	RankIC\	retained
A2	cap	0.06%	0.07	0.12	11.06%		
A2	joint	0.35%	0.64	1.18	59.36%		
A3	raw	-3.60%	-3.42	-6.36	100.00%		
A3	industry	-3.00%	-5.04	-9.37	83.23%		
A3	cap	-3.01%	-3.07	-5.70	83.78%		
A3	joint	-2.52%	-4.58	-8.52	69.95%		

OLS rows are standalone residualization diagnostics. Residualization cannot reclassify Tick executions and does not create alpha. Retention is measured against each role's persisted raw RankIC.

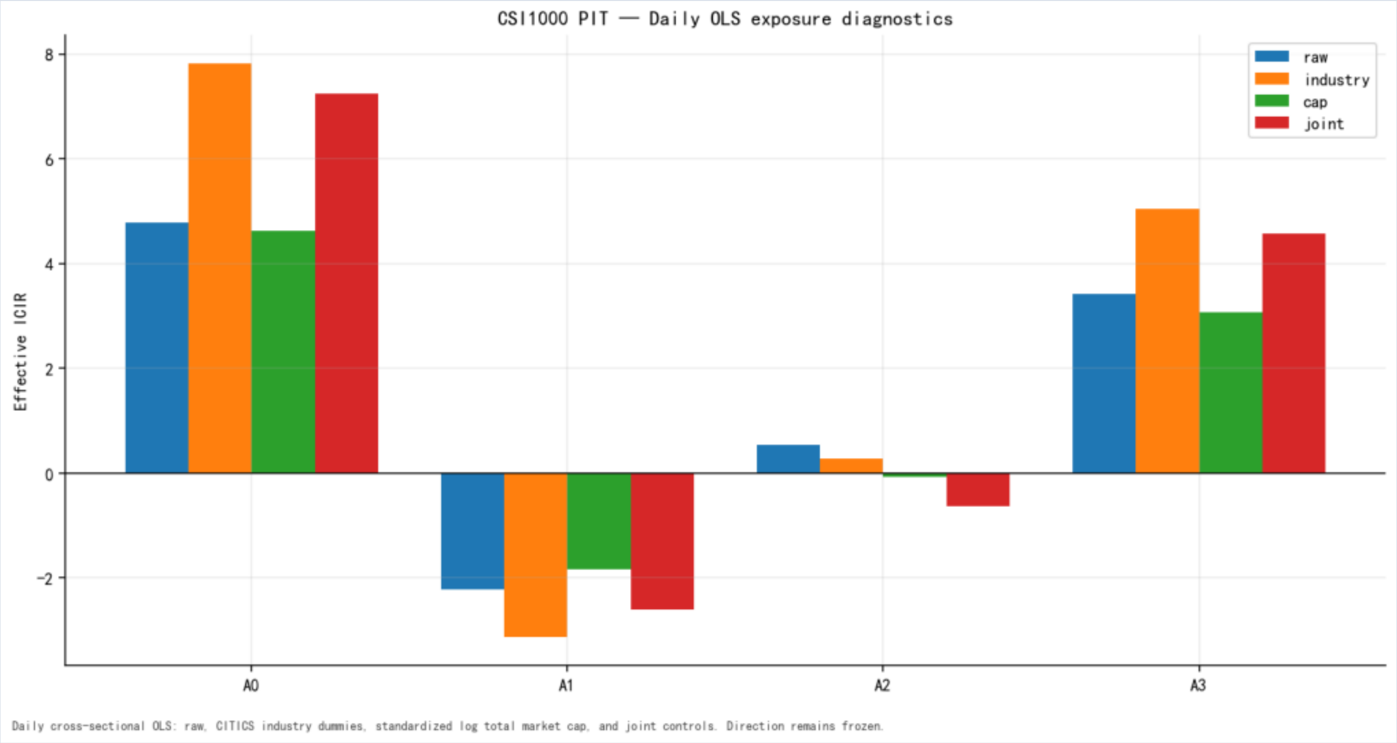
Exposure figures

06_cap_adv_quintiles — Market-cap and ADV quintiles



Market-cap and ADV quintiles — 06_cap_adv_quintiles

09_ols_diagnostics — Raw/industry/cap/joint OLS



Raw/industry/cap/joint OLS — 09_ols_diagnostics

08 — Time and State Robustness

Monthly raw RankIC

Role	Month	RankIC	ICIR	Negative-IC days	Days
A0	2023-01	-3.00%	-6.04	60.00%	15
A0	2023-02	-8.19%	-13.13	85.00%	20
A0	2023-03	-0.89%	-1.16	43.48%	23
A0	2023-04	-3.55%	-3.80	57.89%	19
A0	2023-05	-1.88%	-2.42	55.00%	20
A0	2023-06	-4.27%	-5.09	65.00%	20
A0	2023-07	-8.74%	-16.34	85.71%	21
A0	2023-08	-6.94%	-9.95	78.26%	23
A0	2023-09	-5.99%	-9.55	65.00%	20
A0	2023-10	-4.01%	-7.50	82.35%	17
A0	2023-11	-6.50%	-7.26	72.73%	22
A0	2023-12	-5.29%	-7.26	71.43%	21
A0	2024-01	-5.03%	-6.16	81.82%	22
A0	2024-02	3.10%	2.63	40.00%	15
A0	2024-03	-5.65%	-5.91	61.90%	21

Role	Month	RankIC	ICIR	Negative-IC days	Days
A0	2024-04	-9.39%	-14.52	80.00%	20
A0	2024-05	-7.19%	-16.28	90.00%	20
A0	2024-06	0.02%	0.03	36.84%	19
A0	2024-07	-0.94%	-1.10	60.87%	23
A0	2024-08	-6.35%	-6.61	72.73%	22
A0	2024-09	-0.15%	-0.17	63.16%	19
A0	2024-10	-2.03%	-1.36	61.11%	18
A0	2024-11	-6.30%	-5.86	61.90%	21
A0	2024-12	-6.80%	-7.24	68.18%	22
A0	2025-01	-4.58%	-3.62	66.67%	18
A0	2025-02	2.45%	1.95	44.44%	18
A0	2025-03	-7.94%	-15.60	80.95%	21
A0	2025-04	-6.19%	-8.82	76.19%	21
A0	2025-05	-7.92%	-9.47	73.68%	19
A0	2025-06	-0.46%	-0.65	60.00%	20

Role	Month	RankIC	ICIR	Negative-IC days	Days
A0	2025-07	-4.03%	-5.04	60.87%	23
A0	2025-08	-0.62%	-0.67	57.14%	21
A0	2025-09	-1.01%	-1.11	40.91%	22
A0	2025-10	-7.45%	-9.63	64.71%	17
A0	2025-11	-5.62%	-9.17	70.00%	20
A0	2025-12	-1.76%	-2.41	47.83%	23
A0	2026-01	-3.02%	-3.18	55.00%	20
A0	2026-02	-4.96%	-6.05	64.29%	14
A0	2026-03	-7.02%	-8.98	72.73%	22
A0	2026-04	-1.56%	-2.72	61.90%	21
A0	2026-05	0.92%	0.97	44.44%	18
A0	2026-06	4.31%	3.29	47.62%	21
A0	2026-07	-15.39%	-10.96	78.26%	23
A1	2023-01	5.71%	11.96	33.33%	15
A1	2023-02	3.78%	6.64	30.00%	20

Role	Month	RankIC	ICIR	Negative-IC days	Days
A1	2023-03	0.67%	0.86	47.83%	23
A1	2023-04	0.18%	0.16	57.89%	19
A1	2023-05	3.88%	3.99	35.00%	20
A1	2023-06	-0.46%	-0.44	50.00%	20
A1	2023-07	11.92%	10.91	38.10%	21
A1	2023-08	-1.45%	-1.53	43.48%	23
A1	2023-09	3.67%	4.66	30.00%	20
A1	2023-10	0.69%	0.99	52.94%	17
A1	2023-11	2.47%	3.16	54.55%	22
A1	2023-12	3.17%	2.59	52.38%	21
A1	2024-01	5.67%	6.77	27.27%	22
A1	2024-02	-4.82%	-6.31	53.33%	15
A1	2024-03	-2.98%	-2.80	47.62%	21
A1	2024-04	5.18%	4.83	45.00%	20
A1	2024-05	0.82%	1.21	55.00%	20

Role	Month	RankIC	ICIR	Negative-IC days	Days
A1	2024-06	-2.00%	-2.09	52.63%	19
A1	2024-07	4.55%	3.94	43.48%	23
A1	2024-08	2.52%	3.50	40.91%	22
A1	2024-09	4.14%	7.75	31.58%	19
A1	2024-10	-2.90%	-3.02	61.11%	18
A1	2024-11	3.26%	2.71	38.10%	21
A1	2024-12	5.17%	4.54	31.82%	22
A1	2025-01	0.15%	0.09	44.44%	18
A1	2025-02	0.18%	0.12	50.00%	18
A1	2025-03	7.82%	4.97	47.62%	21
A1	2025-04	1.79%	1.25	57.14%	21
A1	2025-05	6.86%	6.14	31.58%	19
A1	2025-06	2.00%	2.06	50.00%	20
A1	2025-07	5.13%	4.89	34.78%	23
A1	2025-08	-0.10%	-0.10	42.86%	21

Role	Month	RankIC	ICIR	Negative-IC days	Days
A1	2025-09	1.81%	1.11	54.55%	22
A1	2025-10	5.98%	3.21	41.18%	17
A1	2025-11	6.38%	4.67	30.00%	20
A1	2025-12	0.41%	0.37	52.17%	23
A1	2026-01	1.84%	1.47	55.00%	20
A1	2026-02	-4.03%	-2.44	50.00%	14
A1	2026-03	8.51%	6.15	31.82%	22
A1	2026-04	-3.10%	-3.01	61.90%	21
A1	2026-05	-0.72%	-0.48	55.56%	18
A1	2026-06	-4.90%	-2.48	61.90%	21
A1	2026-07	20.34%	8.47	21.74%	23
A2	2023-01	-1.14%	-1.12	53.33%	15
A2	2023-02	-2.30%	-3.04	55.00%	20
A2	2023-03	-2.35%	-2.39	60.87%	23
A2	2023-04	-4.38%	-5.52	63.16%	19

Role	Month	RankIC	ICIR	Negative-IC days	Days
A2	2023-05	-0.60%	-0.50	50.00%	20
A2	2023-06	-0.60%	-0.70	55.00%	20
A2	2023-07	-7.57%	-8.65	76.19%	21
A2	2023-08	1.63%	1.38	56.52%	23
A2	2023-09	-0.76%	-0.84	55.00%	20
A2	2023-10	0.38%	0.37	52.94%	17
A2	2023-11	1.90%	2.04	40.91%	22
A2	2023-12	-0.61%	-0.68	61.90%	21
A2	2024-01	-12.97%	-12.18	81.82%	22
A2	2024-02	4.77%	4.95	40.00%	15
A2	2024-03	0.37%	0.42	52.38%	21
A2	2024-04	2.29%	2.09	35.00%	20
A2	2024-05	1.84%	2.24	40.00%	20
A2	2024-06	-0.37%	-0.27	52.63%	19
A2	2024-07	-5.43%	-4.31	52.17%	23

Role	Month	RankIC	ICIR	Negative-IC days	Days
A2	2024-08	-0.87%	-1.07	50.00%	22
A2	2024-09	0.16%	0.10	47.37%	19
A2	2024-10	-1.97%	-1.85	61.11%	18
A2	2024-11	2.26%	1.76	57.14%	21
A2	2024-12	0.01%	0.01	59.09%	22
A2	2025-01	3.01%	3.21	44.44%	18
A2	2025-02	0.91%	0.94	50.00%	18
A2	2025-03	-0.80%	-0.69	52.38%	21
A2	2025-04	0.44%	0.36	52.38%	21
A2	2025-05	-2.92%	-4.40	52.63%	19
A2	2025-06	1.42%	1.56	55.00%	20
A2	2025-07	-0.14%	-0.13	47.83%	23
A2	2025-08	5.19%	5.58	33.33%	21
A2	2025-09	0.48%	0.40	40.91%	22
A2	2025-10	-2.55%	-2.04	58.82%	17

Role	Month	RankIC	ICIR	Negative-IC days	Days
A2	2025-11	-2.78%	-2.67	55.00%	20
A2	2025-12	1.32%	1.24	47.83%	23
A2	2026-01	0.89%	0.73	45.00%	20
A2	2026-02	0.65%	0.63	57.14%	14
A2	2026-03	-3.84%	-3.28	68.18%	22
A2	2026-04	8.44%	8.48	33.33%	21
A2	2026-05	2.51%	2.65	44.44%	18
A2	2026-06	4.99%	4.49	42.86%	21
A2	2026-07	-11.83%	-7.75	69.57%	23
A3	2023-01	-1.95%	-1.90	60.00%	15
A3	2023-02	-7.75%	-10.02	70.00%	20
A3	2023-03	-4.54%	-4.65	60.87%	23
A3	2023-04	-6.36%	-8.89	73.68%	19
A3	2023-05	-2.99%	-2.65	60.00%	20
A3	2023-06	-3.10%	-3.30	55.00%	20

Role	Month	RankIC	ICIR	Negative-IC days	Days
A3	2023-07	-12.29%	-14.10	85.71%	21
A3	2023-08	-1.75%	-1.46	60.87%	23
A3	2023-09	-3.34%	-3.53	55.00%	20
A3	2023-10	-1.39%	-1.37	64.71%	17
A3	2023-11	-1.63%	-1.74	50.00%	22
A3	2023-12	-2.50%	-2.84	61.90%	21
A3	2024-01	-15.55%	-13.31	86.36%	22
A3	2024-02	2.78%	2.63	33.33%	15
A3	2024-03	-4.26%	-4.05	66.67%	21
A3	2024-04	-1.97%	-1.89	50.00%	20
A3	2024-05	-2.52%	-3.02	50.00%	20
A3	2024-06	-1.86%	-1.22	63.16%	19
A3	2024-07	-6.91%	-5.71	60.87%	23
A3	2024-08	-3.52%	-4.52	63.64%	22
A3	2024-09	-3.10%	-2.11	52.63%	19

Role	Month	RankIC	ICIR	Negative-IC days	Days
A3	2024-10	-7.37%	-5.25	72.22%	18
A3	2024-11	-2.96%	-2.16	61.90%	21
A3	2024-12	-4.71%	-4.41	63.64%	22
A3	2025-01	-0.02%	-0.03	44.44%	18
A3	2025-02	-0.10%	-0.10	50.00%	18
A3	2025-03	-5.84%	-5.72	57.14%	21
A3	2025-04	-4.91%	-5.12	61.90%	21
A3	2025-05	-8.31%	-11.22	68.42%	19
A3	2025-06	-0.75%	-0.77	55.00%	20
A3	2025-07	-4.78%	-4.37	52.17%	23
A3	2025-08	1.47%	1.51	47.62%	21
A3	2025-09	-2.00%	-1.82	59.09%	22
A3	2025-10	-7.08%	-5.91	64.71%	17
A3	2025-11	-4.90%	-5.00	60.00%	20
A3	2025-12	-0.44%	-0.43	52.17%	23

Role	Month	RankIC	ICIR	Negative-IC days	Days
A3	2026-01	-0.59%	-0.50	40.00%	20
A3	2026-02	-2.05%	-2.02	57.14%	14
A3	2026-03	-5.32%	-5.30	72.73%	22
A3	2026-04	5.34%	5.78	38.10%	21
A3	2026-05	0.40%	0.47	44.44%	18
A3	2026-06	2.15%	2.15	38.10%	21
A3	2026-07	-10.87%	-9.96	73.91%	23

Rolling 63-trading-day RankIC

Role	Latest date	Latest 63d	Minimum 63d	Maximum 63d	Frozen-direction windows	Latest count
A0	2026-07-31	-3.89%	-8.55%	0.54%	99.76%	63
A1	2026-07-31	5.50%	-2.35%	6.80%	6.39%	63
A2	2026-07-31	-1.55%	-4.80%	5.19%	57.51%	63
A3	2026-07-31	-2.82%	-7.95%	2.49%	95.15%	63

The 63-day table is derived only from persisted rolling observations. Its direction share is the fraction of finite windows matching frozen direction -1.

Lagged-turnover terciles

Role	Turnover tercile	RankIC	ICIR	Negative-IC days	Days
A0	Low	-1.25%	-2.05	55.85%	795
A0	Mid	-2.46%	-4.21	61.01%	795
A0	High	-4.77%	-6.04	65.41%	795
A1	Low	-0.28%	-0.38	52.08%	795
A1	Mid	-0.41%	-0.48	53.08%	795
A1	High	1.44%	1.41	46.67%	795
A2	Low	-0.69%	-0.58	51.32%	795
A2	Mid	-0.37%	-0.32	51.07%	795
A2	High	1.52%	1.58	46.29%	795
A3	Low	-2.22%	-1.98	54.59%	795
A3	Mid	-2.42%	-2.16	55.35%	795
A3	High	-3.32%	-3.70	59.87%	795

Low/Mid/High states use lagged turnover information. The direction is not re-fitted within a state.

Frozen IS / validation / OOS

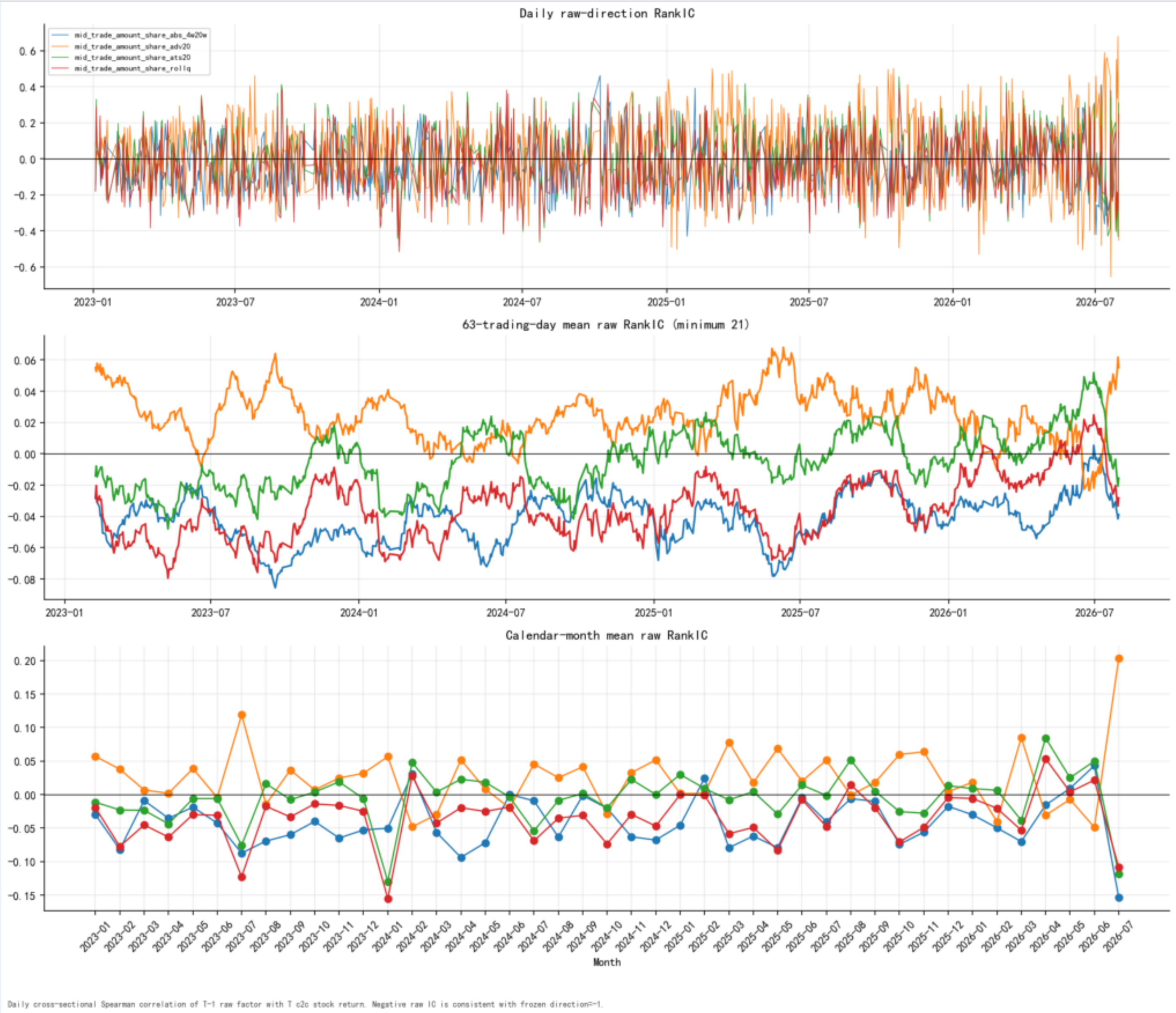
Role	Segment	Actual dates	RankIC	t-stat	ICIR	Sharpe	MDD	Turnover
A0	IS	2023-01-04 to 2024-06-28	-4.80%	-7.58	-6.33	2.74	-9.68%	1.69
A0	validation	2023-07-03 to 2024-06-28	-5.39%	-7.00	-7.13	3.46	-9.68%	1.66
A0	OOS	2024-07-01 to 2026-07-31	-3.84%	-5.67	-3.98	0.85	-30.12%	1.59
A1	IS	2023-01-04 to 2024-06-28	2.08%	2.69	2.25	-0.33	-22.31%	1.19
A1	validation	2023-07-03 to 2024-06-28	2.05%	2.12	2.16	-0.51	-17.49%	1.20
A1	OOS	2024-07-01 to 2026-07-31	3.15%	3.22	2.26	-0.71	-44.28%	1.11
A2	IS	2023-01-04 to 2024-06-28	-1.27%	-1.53	-1.28	0.15	-16.48%	1.04
A2	validation	2023-07-03 to 2024-06-28	-0.96%	-0.91	-0.93	-0.03	-16.48%	1.05
A2	OOS	2024-07-01 to 2026-07-31	-0.09%	-0.12	-0.08	-0.87	-62.82%	1.15
A3	IS	2023-01-04 to 2024-06-28	-4.26%	-4.95	-4.13	1.64	-12.09%	1.12

Role	Segment	Actual dates	RankIC	t-stat	ICIR	Sharpe	MDD	Turnover
A3	validation	2023-07-03 to 2024-06-28	-4.12%	-3.75	-3.82	1.53	-12.09%	1.14
A3	OOS	2024-07-01 to 2026-07-31	-3.13%	-4.18	-2.94	0.26	-40.16%	1.22

IS, validation and OOS are shown separately. Parameters and direction remain frozen in every segment; OOS raw RankIC and t-stat, rather than the segment's highest Sharpe, drive the final decision.

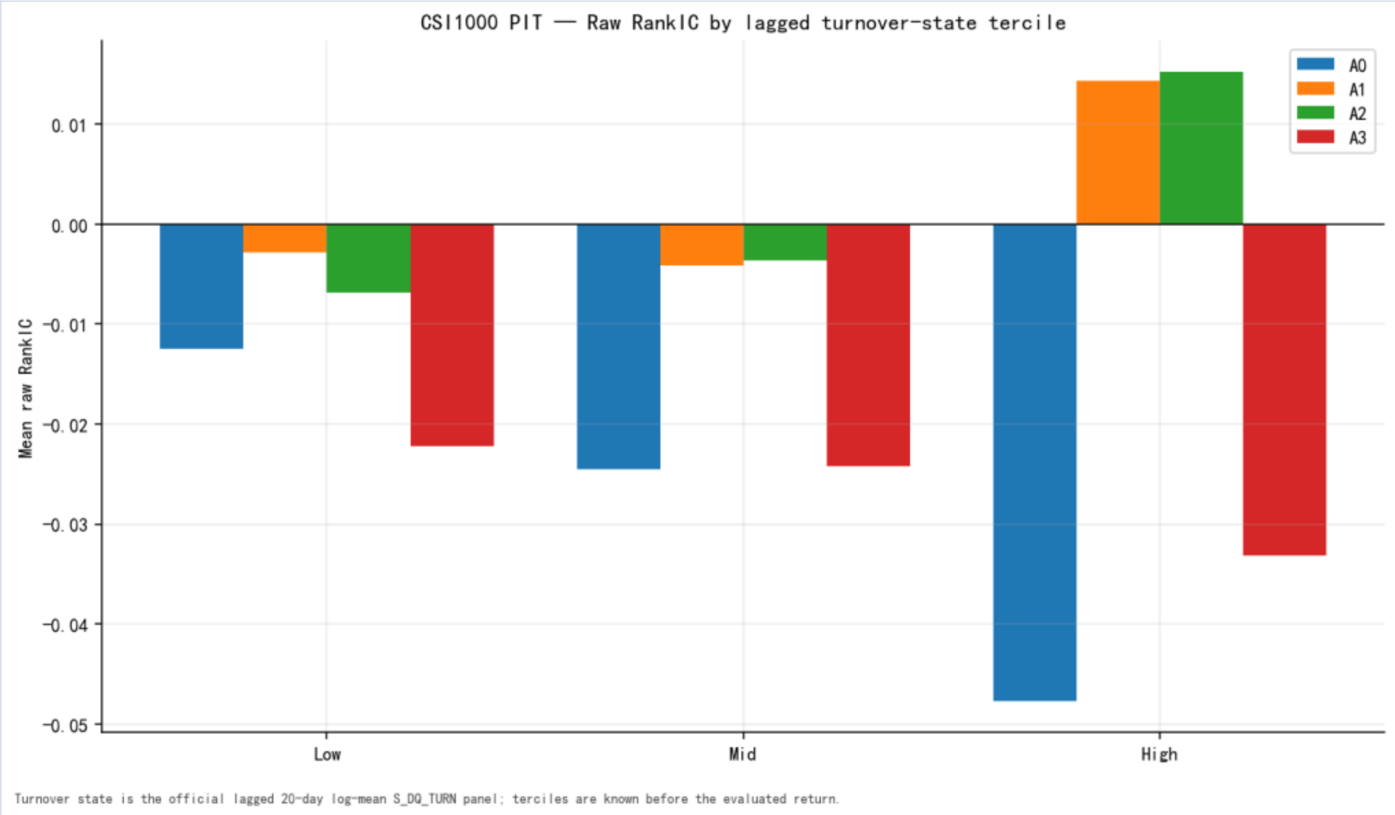
Time, state, segment and coverage figures

05_ic_stability — Daily, monthly, and 63-day RankIC



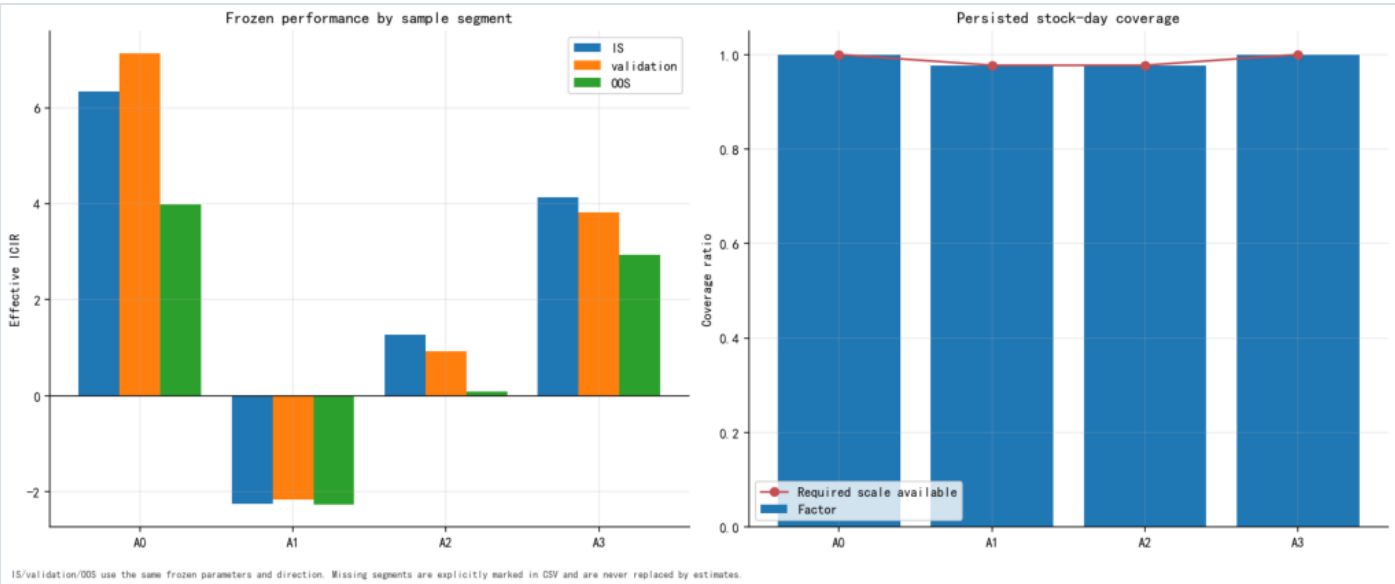
Daily, monthly, and 63-day RankIC — 05_ic_stability

08_turnover_tercile — Lagged turnover-state terciles



Lagged turnover-state terciles — 08_turnover_tercile

10_segments_and_coverage — IS/validation/OOS and coverage



IS/validation/OOS and coverage — 10_segments_and_coverage

09 — In-sample / Out-of-sample Protocol

Frozen chronology

- Distribution-only calibration: 2023-01-03 to 2023-06-30.
- Formal-sample comparability: return dates 2023-01-04 to 2024-06-28.
- Post-calibration validation: return dates 2023-07-03 to 2024-06-28.
- True OOS target: 2024-07-01 to the latest common complete month across

strict Tick, Wind returns, tradability data, and PIT membership.

The target OOS endpoint is 2026-07-31 when all sources pass coverage checks.

Frozen objects

Before any return evaluation, frozen_config.json records:

- A1 lower and upper bps-of-ADV bounds;
- A2 0.5–2.0 bounds;
- candidate grids and the distribution-only A1 selection rule;
- effective direction -1;
- calibration dates;
- strict filter and source-cache hashes.

Stage B verifies the config checksum and refuses to continue if it changes. Validation and OOS periods cannot select thresholds or infer direction again.

Interpretation

The formal-sample table preserves comparability with the previous report. Because A1 is selected without return information, it is not return-optimized; nevertheless the post-calibration and true OOS tables are the stronger evidence. IS and OOS are reported separately and are never concatenated for parameter selection.

Execution limitation

The factor is complete only after the T-1 close. T-1 signal versus T close-to-close return includes the overnight component and is not a complete execution simulation. H-L results are fee-zero standalone sorting diagnostics, not portfolio-strategy returns.

10 — Research Decision

Final branch

only the fixed-RMB phenomenon remains

Frozen decision evidence

Role	OOS raw RankIC	OOS t-stat	Frozen direction	Significant	Pools same direction	Retained
A0	-3.84%	-5.67	yes	yes	4/4	yes
A1	3.15%	3.22	no	no	0/4	no
A2	-0.09%	-0.12	yes	no	3/4	no
A3	-3.13%	-4.18	yes	yes	4/4	yes

A role is retained only if all three primary gates pass:

1. CSI1000 OOS raw RankIC has frozen direction -1;
2. its direction-adjusted raw RankIC t-stat is at least 1.96;
3. raw RankIC has the frozen direction in ALL, CSI300, CSI500 and CSI1000.

The four-pool evidence is the persisted full-common-sample universe table; the persisted segment table supplies CSI1000 OOS significance. No segment-by-pool result is invented.

The branch order is fixed: both A1/A2, A1 only, A2 only, A0-only fallback, then no retained normalized relation. A3 remains a secondary P1 diagnostic and cannot select a branch.

Parameter-stability evidence

Role	Candidates same direction	Share	Selected RankIC	Selected t-stat	Selected ICIR
A0	1/1	100.00%	-4.24%	-8.91	-4.79
A1	1/10	10.00%	2.70%	4.13	2.22
A2	8/10	80.00%	-0.58%	-1.01	-0.54
A3	1/1	100.00%	-3.60%	-6.36	-3.42

Parameter stability is supporting evidence after parameters were frozen. It cannot overturn failed OOS direction/significance gates and cannot select the highest Sharpe.

Cost and interpretation limits

- the cost label is one-way 7.5 bps, display-only;

- H-L is a gross standalone sorting diagnostic, not a portfolio strategy;
- Tick execution amount does not identify investor type;
- no causal claim follows from OLS residualization.

Explicitly outside scope

This report does not perform factor-library correlation analysis, factor combination, incremental-IC testing, portfolio optimization, alpha stacking, or return-based parameter optimization.

Code Evidence Map

This map freezes the implementation evidence used by the normalized single-factor study. It distinguishes the authoritative strict construction from deprecated legacy artifacts.

Strict Tick construction

- l2_factor_reproduction/python/ch_tick.py::_regular_session_filter_sql

applies the regular-session predicate independently to every Tick row.

- l2_factor_reproduction/python/ch_tick.py::fetch_tick_bucketed

defines SSE executions as Type='T' with amount=ifNull(Amount, Price*Volume).

- The same function defines SZSE executions as Type='011' with

BidOrderNo>0, AskOrderNo>0, and amount=Price*Volume.

- Both numerator and denominator are traded amounts. Tick rows are

executions, not reconstructed parent orders.

Authoritative A0

- Strict cache:

research/results/l2_reproduction/mid_order_ratio/analysis/param_sensitivity/tick_bucketed_strict_trade_2023-01-01_2024-06-30.parquet

- Strict-cache metadata:

research/results/l2_reproduction/mid_order_ratio/analysis/param_sensitivity/tick_bucketed_strict_trade_2023-01-01_2024-06-30.metadata.json

- Frozen checksum:

ccd2f23475756052c60c32f8b56b8cbb648ab99508bf866c9b2424b7ff61cb1f

- Formula:

(cum_200000-cum_40000)/TotalAmount

- Precise factor id:

mid_trade_amount_share_abs_4w20w

- Legacy alias:

mid_order_ratio

The normalized package runs run_mid_trade_amount_normalized_parity.py before any return evaluation. On the authoritative cache date grid it uses the canonical values exactly and sets dynamic-only keys to missing. After 2024-06-28 it extends A0 from the same strict daily amount primitive only after full-period dynamic-versus- primitive parity passes.

Normalized dynamic construction

- ch_mid_trade_amount_normalization.py::build_dynamic_factor_sql joins lagged ADV20_lag1 / ATS20_lag1 and same-day Q20/Q80 through clickhouse-connect ExternalData.

- A0, all nine A1 candidates, all nine A2 candidates, and A3 are aggregated

in one strict Tick scan per exchange and month. Zero-selected-amount groups are represented by numeric zero; missing historical scales are restored to factor NaN by the scale-evidence gate in build_mid_trade_amount_normalized_cache.py::build_factor_panels.

- The cache builder enforces unique keys, exact scale joins, nonnegative

selected amounts, selected-amount bounds, total-amount conservation, frozen config SHA256, resumable chunk request hashes, and a two-worker default with a hard maximum of ten.

- finalize_mid_trade_amount_normalized_lineage.py stores each chunk's exact

SSE/SZSE SQL text, SQL SHA256, strict filters, ExternalData format, client version, server version, artifact hash, dates, row count, and parent lineage.

Evaluation contract

- generate_mid_order_ratio_report_artifacts.py::align_core_panels applies

factor-date PIT/tradability masks before the one-trading-day signal lag.

- generate_mid_order_ratio_report_artifacts.py::evaluate_prepared computes

daily Spearman RankIC, ICIR with ddof=1, equal-weight deciles, H-L, turnover, and frozen-direction diagnostics.

- Factor_Dev_Lib.py::groupTest defines H-L=G10-G1.
- Factor_Dev_Lib.py::implied_annu_fee defines the display-only fee as

daily H-L turnover times 7.5 bps times 250.

- The frozen effective direction is -1; no evaluated return window may

infer or change it.

- generate_mid_trade_amount_normalized_report.py reuses these functions for

all four PIT universes and all IS/validation/OOS segments. Its A0 input is required to be the parity-gated authoritative panel; the dynamic A0 copy cannot silently become the headline source.

PIT and exposure data

- Factor_Dev_Lib.py::get_index_member_mask supplies Wind point-in-time

index membership.

- Factor_Dev_Lib.py::get_EOD_Not_Limit, get_EOD_Not_ST, and

get_TradeStatus supply the common tradability mask.

- Factor_Dev_Lib.py::panel_neutral_size_ind and

cs_neutral_size_ind implement daily industry-dummy and log-market-cap OLS residualization.

- Industry data are CITICS level-1 classifications; market cap is Wind

S_VAL_MV, transformed with log, MAD treatment, and cross-sectional z-score.

Legacy exclusion

research/results/l2_reproduction/mid_order_ratio/LEGACY_QUERY_WARNING.md states that pre-2026-08-04 canonical/neutralization artifacts are not formal research evidence. This study never reads the legacy factor_narrow.parquet as an A0 source.

Dynamic Factor SQL Evidence

This is the exact first monthly query pair from the persisted dynamic cache. Every monthly chunk metadata file stores its own SSE and SZSE SQL text and SHA256. The execution used clickhouse-connect 0.8.17 against ClickHouse 23.2.3.17 with CSVWithNames ExternalData.

Chunk: 2023-01-03 to 2023-01-31.

SSE

```
WITH strict_ticks AS
    (

    SELECT
        Symbol,
        ExchTime,
        toFloat64(ifNull(Amount, Price * Volume)) AS amt
    FROM cmds.SSE_AL_TICK_EXG
    WHERE ExchTime >= toDateTime64('2023-01-03 09:30:00', 6, 'Asia/Shanghai')
      AND ExchTime <  toDateTime64('2023-01-31 15:00:01', 6, 'Asia/Shanghai')
      AND toDate(ExchTime) BETWEEN toDate('2023-01-03') AND toDate('2023-01-31')
      AND ((toHour(ExchTime) = 9 AND toMinute(ExchTime) >= 30) OR (toHour(ExchTime) > 9 AND toHour(ExchTime) <
15) OR (toHour(ExchTime) = 15 AND toMinute(ExchTime) = 0 AND toSecond(ExchTime) = 0))
      AND Type = 'T'
      AND Price > 0 AND Volume > 0
      AND startsWith(Symbol, '6')


    ),
    joined_ticks AS
    (
        SELECT
            t.Symbol AS Symbol,
            toDate(t.ExchTime) AS TradeDate,
            t.amt AS amt,
            s.ADV20_lag1 AS ADV20_lag1,
            s.ATS20_lag1 AS ATS20_lag1,
            s.q20 AS q20,
            s.q80 AS q80
        FROM strict_ticks AS t
        INNER JOIN scale_rows AS s
            ON t.Symbol = s.symbol
           AND toDate(t.ExchTime) = s.TradeDate
    )
    SELECT
        concat(Symbol, '.SH') AS symbol,
        TradeDate,
        sum(amt) AS total_amount,
        ifNull(sumIf(amt, amt > 40000 AND amt <= 200000), 0) AS `a0_abs_4w20w_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 0.5 AND (amt
/ ADV20_lag1 * 10000) <= 5), 0) AS `a1_adv20_bps_l0p5_h5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 0.5 AND (amt
/ ADV20_lag1 * 10000) <= 10), 0) AS `a1_adv20_bps_l0p5_h10_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 0.5 AND (amt
/ ADV20_lag1 * 10000) <= 20), 0) AS `a1_adv20_bps_l0p5_h20_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 1 AND (amt /
ADV20_lag1 * 10000) <= 5), 0) AS `a1_adv20_bps_l1_h5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 1 AND (amt /
ADV20_lag1 * 10000) <= 10), 0) AS `a1_adv20_bps_l1_h10_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 1 AND (amt /
ADV20_lag1 * 10000) <= 20), 0) AS `a1_adv20_bps_l1_h20_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 2 AND (amt /
ADV20_lag1 * 10000) <= 5), 0) AS `a1_adv20_bps_l2_h5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 2 AND (amt /
ADV20_lag1 * 10000) <= 10), 0) AS `a1_adv20_bps_l2_h10_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 2 AND (amt /
ADV20_lag1 * 10000) <= 20), 0) AS `a1_adv20_bps_l2_h20_selected_amount`,
```

```
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.25 AND (amt /
ATS20_lag1) <= 1.5), 0) AS `a2_ats20_l0p25_h1p5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.25 AND (amt /
ATS20_lag1) <= 2), 0) AS `a2_ats20_l0p25_h2_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.25 AND (amt /
ATS20_lag1) <= 3), 0) AS `a2_ats20_l0p25_h3_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.5 AND (amt /
ATS20_lag1) <= 1.5), 0) AS `a2_ats20_l0p5_h1p5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.5 AND (amt /
ATS20_lag1) <= 2), 0) AS `a2_ats20_l0p5_h2_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.5 AND (amt /
ATS20_lag1) <= 3), 0) AS `a2_ats20_l0p5_h3_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.75 AND (amt /
ATS20_lag1) <= 1.5), 0) AS `a2_ats20_l0p75_h1p5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.75 AND (amt /
ATS20_lag1) <= 2), 0) AS `a2_ats20_l0p75_h2_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.75 AND (amt /
ATS20_lag1) <= 3), 0) AS `a2_ats20_l0p75_h3_selected_amount`,
        ifNull(sumIf(amt, isNotNull(q20) AND isNotNull(q80) AND q20 >= 0 AND q80 >= q20 AND amt > q20 AND amt <=
q80), 0) AS `a3_q20_q80_selected_amount`
    FROM joined_ticks
    GROUP BY Symbol, TradeDate
    ORDER BY TradeDate, symbol
```

SZSE

```
WITH strict_ticks AS
    (

    SELECT
        Symbol,
        ExchTime,
        toFloat64(Price * Volume) AS amt
    FROM cmds.SZSE_AL_TICK_EXG
    WHERE ExchTime >= toDateTime64('2023-01-03 09:30:00', 6, 'Asia/Shanghai')
      AND ExchTime <  toDateTime64('2023-01-31 15:00:01', 6, 'Asia/Shanghai')
      AND toDate(ExchTime) BETWEEN toDate('2023-01-03') AND toDate('2023-01-31')
      AND ((toHour(ExchTime) = 9 AND toMinute(ExchTime) >= 30) OR (toHour(ExchTime) > 9 AND toHour(ExchTime) <
15) OR (toHour(ExchTime) = 15 AND toMinute(ExchTime) = 0 AND toSecond(ExchTime) = 0))
      AND Type = '011' AND BidOrderNo > 0 AND AskOrderNo > 0
      AND Price > 0 AND Volume > 0
      AND (startsWith(Symbol, '000') OR startsWith(Symbol, '001') OR startsWith(Symbol, '002') OR
startsWith(Symbol, '003') OR startsWith(Symbol, '300') OR startsWith(Symbol, '301') OR startsWith(Symbol, '302'))



    ),
    joined_ticks AS
    (
        SELECT
            t.Symbol AS Symbol,
            toDate(t.ExchTime) AS TradeDate,
            t.amt AS amt,
            s.ADV20_lag1 AS ADV20_lag1,
            s.ATS20_lag1 AS ATS20_lag1,
            s.q20 AS q20,
            s.q80 AS q80
        FROM strict_ticks AS t
        INNER JOIN scale_rows AS s
            ON t.Symbol = s.symbol
           AND toDate(t.ExchTime) = s.TradeDate
    )
    SELECT
        concat(Symbol, '.SZ') AS symbol,
        TradeDate,
        sum(amt) AS total_amount,
        ifNull(sumIf(amt, amt > 40000 AND amt <= 200000), 0) AS `a0_abs_4w20w_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 0.5 AND (amt
/ ADV20_lag1 * 10000) <= 5), 0) AS `a1_adv20_bps_l0p5_h5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 0.5 AND (amt
```

```
/ ADV20_lag1 * 10000) <= 10), 0) AS `a1_adv20_bps_l0p5_h10_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 0.5 AND (amt
/ ADV20_lag1 * 10000) <= 20), 0) AS `a1_adv20_bps_l0p5_h20_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 1 AND (amt /
ADV20_lag1 * 10000) <= 5), 0) AS `a1_adv20_bps_l1_h5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 1 AND (amt /
ADV20_lag1 * 10000) <= 10), 0) AS `a1_adv20_bps_l1_h10_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 1 AND (amt /
ADV20_lag1 * 10000) <= 20), 0) AS `a1_adv20_bps_l1_h20_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 2 AND (amt /
ADV20_lag1 * 10000) <= 5), 0) AS `a1_adv20_bps_l2_h5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 2 AND (amt /
ADV20_lag1 * 10000) <= 10), 0) AS `a1_adv20_bps_l2_h10_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ADV20_lag1) AND ADV20_lag1 > 0 AND (amt / ADV20_lag1 * 10000) > 2 AND (amt /
ADV20_lag1 * 10000) <= 20), 0) AS `a1_adv20_bps_l2_h20_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.25 AND (amt /
ATS20_lag1) <= 1.5), 0) AS `a2_ats20_l0p25_h1p5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.25 AND (amt /
ATS20_lag1) <= 2), 0) AS `a2_ats20_l0p25_h2_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.25 AND (amt /
ATS20_lag1) <= 3), 0) AS `a2_ats20_l0p25_h3_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.5 AND (amt /
ATS20_lag1) <= 1.5), 0) AS `a2_ats20_l0p5_h1p5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.5 AND (amt /
ATS20_lag1) <= 2), 0) AS `a2_ats20_l0p5_h2_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.5 AND (amt /
ATS20_lag1) <= 3), 0) AS `a2_ats20_l0p5_h3_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.75 AND (amt /
ATS20_lag1) <= 1.5), 0) AS `a2_ats20_l0p75_h1p5_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.75 AND (amt /
ATS20_lag1) <= 2), 0) AS `a2_ats20_l0p75_h2_selected_amount`,
        ifNull(sumIf(amt, isNotNull(ATS20_lag1) AND ATS20_lag1 > 0 AND (amt / ATS20_lag1) > 0.75 AND (amt /
ATS20_lag1) <= 3), 0) AS `a2_ats20_l0p75_h3_selected_amount`,
        ifNull(sumIf(amt, isNotNull(q20) AND isNotNull(q80) AND q20 >= 0 AND q80 >= q20 AND amt > q20 AND amt <=
q80), 0) AS `a3_q20_q80_selected_amount`
    FROM joined_ticks
    GROUP BY Symbol, TradeDate
    ORDER BY TradeDate, symbol
```

Appendix — Reproduction Commands

Run from the repository root with the research environment:

```
PY=/opt/conda/anaconda3/envs/base_93/bin/python

# 1. Build resumable strict daily primitives and lagged scales.
$PY l2_factor_reproduction/scripts/build_mid_trade_amount_normalized_cache.py \
  --stage primitives

# 2. Distribution-only calibration; this stage must not load returns.
$PY l2_factor_reproduction/scripts/build_mid_trade_amount_normalized_cache.py \
  --stage calibrate

# 3. Refuse to run unless frozen_config.json passes its hash checks.
$PY l2_factor_reproduction/scripts/build_mid_trade_amount_normalized_cache.py \
  --stage factors --workers 2

# 4. Enrich every resumable chunk with exact SQL/runtime lineage.
$PY l2_factor_reproduction/scripts/finalize_mid_trade_amount_normalized_lineage.py

# 5. Run the A0 panel hard gate and materialize the authoritative headline.
$PY l2_factor_reproduction/scripts/run_mid_trade_amount_normalized_parity.py \
  --stage panel

# 6. Generate standalone artifacts and all ten figure classes.
$PY l2_factor_reproduction/scripts/generate_mid_trade_amount_normalized_report.py

# 7. Render the six numeric Markdown chapters from persisted artifacts.
$PY l2_factor_reproduction/reporting/render_mid_trade_amount_normalized_markdown.py

# 8. Recheck A0 formal RankIC/ICIR/decile/H-L parity.
$PY l2_factor_reproduction/scripts/run_mid_trade_amount_normalized_parity.py \
  --stage formal

# 9. Export Markdown to embedded HTML and PDF, then hash the package.
$PY l2_factor_reproduction/scripts/export_mid_trade_amount_normalized_report.py
$PY l2_factor_reproduction/scripts/finalize_mid_trade_amount_normalized_package.py

# 10. Unit, regression, artifact, and package gates.
$PY -m pytest -q \
  l2_factor_reproduction/tests/test_mid_trade_amount_normalization.py \
  l2_factor_reproduction/tests/test_ch_mid_trade_amount_normalization.py \
  l2_factor_reproduction/tests/test_build_mid_trade_amount_normalized_cache.py \
  l2_factor_reproduction/tests/test_mid_trade_amount_normalized_report.py \
  l2_factor_reproduction/tests/test_render_mid_trade_amount_normalized_markdown.py \
  l2_factor_reproduction/tests/test_mid_trade_amount_normalized_artifacts.py \
  l2_factor_reproduction/tests/test_standalone_report_export.py \
  l2_factor_reproduction/tests/test_ch_tick.py \
  l2_factor_reproduction/tests/test_mid_order_ratio_report_artifacts.py \
  l2_factor_reproduction/tests/test_factor_dev_lib_metrics.py
```

The cache builder uses monthly resumable chunks. Existing chunks are reused unless an explicit force flag is supplied. Stage B never changes the frozen thresholds or direction.